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Spatially Periodic Computation in the Entorhinal-Hippocampal Circuit During Navigation

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eLife Assessment

This study offers **important** insights into how entorhinal and hippocampal activity support human thinking in feature spaces. It replicates hexagonal symmetry in entorhinal cortex, reports a novel three-fold symmetry in both behavior and hippocampal signals, and links these findings with a computational model. The task and analyses are sophisticated, and the results appear **solid** and of broad interest to neuroscientists.

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Abstract

To enable navigation in both physical and mental spaces, the human brain employs a cognitive map constructed from the global metrics of the entorhinal cortex and the local representations of the hippocampus. However, how these two regions coordinate to enable navigation remains poorly understood. Here, we designed an object-matching task where human participants unknowingly manipulated object variants arranged in a ring-like structure around a central prototype. Functional MRI revealed a 3-fold spatial periodicity in the hippocampal activity that tracked navigation directions from object variants to the central prototype. This hippocampal periodicity was phase-locked with the well-documented 6-fold periodicity of the entorhinal cortex, suggesting hierarchical interaction between these regions. Consistent with this neural pattern, a corresponding 3-fold periodicity was observed in behavioral performance, which was synchronized with hippocampal activity. Finally, an EC-HPC PhaseSync model reproduced this phenomenon, in which the 6-fold activity periodicity of entorhinal grid cells across directions projects vectorial representations to the hippocampus, and the collection of these vectors exhibits 3-fold periodicity to represent conceptual directions. Together, these findings reveal a periodic mechanism through which entorhinal grid codes structure hippocampal vector representations.

Introduction

The cognitive map, initially introduced by Tolman (Tolman, 1948) and later supported by the discovery of place cells in the hippocampus (HPC) (O'Keefe and Dostrovsky, 1971), has served as a framework for spatial navigation. Grid cells in the upstream entorhinal cortex (EC), known for their hexagonal firing pattern to provide the metric for space (Hafting et al., 2005), have also been found to represent conceptual space beyond the physical reference frame (Constantinescu et al., 2016; Bao et al., 2019; Raithel et al., 2023). These findings suggest that the EC-HPC circuit fundamentally organizes spatial and non-spatial knowledge (Epstein et al., 2017; Behrens et al., 2018; Bottini and Doeller, 2020; Park et al., 2020), and to guide flexible behaviors such as retrieving knowledge from the past and making decisions for the future (Addis et al., 2007; Hassabis and Maguire, 2007; Schacter et al., 2012).

Flexible navigation requires the mental simulation of prospective pathways on a cognitive map, which is defined by essential cues such as self-location, goal location, direction, and distance (Nyberg et al., 2022 [DOI](#)). This process necessarily involves dynamic representations of the external world relative to self-location. Converging evidence indicates that the HPC binds spatial cues into vectorial representations, providing activity gradients that reflect potential pathways toward goals. In bats, a subpopulation of CA1 neurons exhibited conjunctive tuning to both direction and distance, with activity gradually decaying within 10–15 minutes after the goal is displaced (Sarel et al., 2017 [DOI](#)). In rats, similar findings were observed for neurons tuned to featureless goal locations distributed across space (Ormond and O’Keefe, 2022 [DOI](#)). These neurons fire maximally when the animal is oriented toward the goal and rapidly reorganize following goal shifts. Consistently, the neural representation of prospective pathways in the human brain is affected by goal locations (Muhle-Karbe et al., 2023 [DOI](#)). The neural geometry derived from the BOLD signals in the HPC is distorted by goals, reflecting the successful learning of goal-directed navigation. Collectively, these findings suggest that the formation of prospective pathways relies on memory-based vectorial representations in the HPC. However, these studies raise the question of how grid cells in the EC contribute to this process.

We hypothesize that projections from EC grid cell populations provide a coherent cognitive-map framework in the HPC that embeds a three-fold periodic structure across spatial directions to support vectorial representations and simulating the prospective pathways. Grid cells exhibit hexagonal firing patterns with nearly invariant orientations (Hafting et al., 2005 [DOI](#); Sargolini et al., 2006 [DOI](#); Krupic et al., 2012 [DOI](#); Gardner et al., 2022 [DOI](#)). These properties may support the formation of vector-like representations of pathways by coactivation of grid cell populations. In the simplest case, when one mentally simulates a straight pathway aligned with the grid orientation, a subpopulation of grid cells would be sequentially activated, and the resulting population activity would manifest a near-perfect vectorial representation with constant activity strength along the pathway. In contrast, when the pathway is misaligned with the grid orientation, the corresponding grid cell population yields a distorted vectorial code. Consequently, simulating straight pathways spanning 0°–360° yields only half the number of unique activity patterns. This arises because the hexagonal grid’s 180° rotational symmetry makes orientations separated by 180° indistinguishable. We therefore speculate that the vectorial representations embedded in grid cell activity are periodic across spatial orientations and are transmitted through the EC–HPC circuit to bind prospective path directions. Consistent with this idea, reorientation paradigms in both rodents and young children demonstrate that subjects search equally at two opposite directions, reflecting successful orientation encoding but a failure to integrate spatial direction (Hermer and Spelke, 1994 [DOI](#); Julian et al., 2015 [DOI](#); Gallistel, 2017 [DOI](#); Julian et al., 2018 [DOI](#)).

This hypothesis is supported by evidence from anatomical, functional, computational, and physiological findings on the EC–HPC circuit. Anatomically, the EC serves as one of the major sources of input to the downstream HPC (Witter and Amaral, 1991 [DOI](#); van Groen et al., 2003 [DOI](#); Garcia and Buffalo, 2020 [DOI](#)). Functionally, grid cells in the medial EC exhibited multiplexed and heterogeneous responses corresponding to position, direction, and speed before they are integrated in the HPC (Sargolini et al., 2006 [DOI](#); Hardcastle et al., 2017 [DOI](#)). Computationally, grid cells have been proposed as the foundation of hippocampal place field formation by integrating multiple grid modules (Solstad et al., 2006 [DOI](#); De Almeida et al., 2009 [DOI](#); Bush et al., 2014 [DOI](#); Bush et al., 2015 [DOI](#); Bicanski and Burgess, 2019 [DOI](#)). Physiologically, EC lesions disrupt the precision and stability of place fields (Hales et al., 2014 [DOI](#)), leading to reduced discharge rates and field sizes (Van Cauter et al., 2008 [DOI](#)). If our hypothesis is correct, a 3-fold periodicity aligned with the three principal grid axes should emerge in the HPC, phase-locked with EC activity along path directions. Because simultaneous population-level recordings from the EC and HPC remain technically challenging, we employed fMRI, which has previously revealed 6-fold periodicity in the EC (Doeller et al., 2010 [DOI](#); Constantinescu et al., 2016 [DOI](#); Bao et al., 2019 [DOI](#); Wagner et al., 2023 [DOI](#); Raithel et al., 2023 [DOI](#)), to test for periodicity in the HPC.

A novel 3D object, named Greeble (Fig. 1a [↗](#)) (Gauthier and Tarr, 1997 [↗](#)), was used to create a conceptual Greeble space. Within this space, locations were represented by Greeble variants characterized by two features (“Loogit” and “Vacso”). The feature length defined the two dimensions of the space. The central Greeble served as the prototype. Participants were instructed to morph Greeble variants to match this target prototype (Fig. 1b [↗](#)). This process generated a sequence of Greebles that resembled movements along navigational path in a two-dimensional conceptual space (Fig. 1c [↗](#)), although participants were unaware of the underlying Greeble space. To ensure a comprehensive exploration of the Greeble space for detecting hippocampal periodicity, Greeble variants were pseudo-randomly sampled at the periphery of space. This ensured a high-resolution sampling of conceptual directions ranging from 0° to 360° (i.e., the orange locations in Fig. 1c [↗](#)), while controlling for distance. As a result, we observed a 3-fold periodicity in hippocampal activity, cross-validated using sinusoidal modulation and spectral analyses. The spatial phase of the HPC was coupled with a 6-fold periodicity in the EC, no spatial offset was identified. In addition, we identified a 3-fold periodicity in participants’ behavioral performance that was phase-locked with hippocampal activity. Finally, the EC–HPC PhaseSync model, developed to simulate EC projections into the HPC, reproduced the emergence of 3-fold activity periodicity across directions under randomized goal locations. Together, these empirical findings highlight a periodic representation of conceptual directions within the HPC, suggesting that vectorial representations in the hippocampus may arise through projections from periodic grid codes in the EC.

Results

Behavioral results

Participants adjusted Greeble features using two response boxes inside the MRI scanner. Each response box controlled one feature, with two buttons used to stretch and shrink the feature, respectively. To prevent a “horizontal–vertical movement” strategy, in which participants navigate primarily along the cardinal directions (e.g., repeatedly adjusting one feature towards North and then East), participants were encouraged to adjust both features simultaneously to promote directional variability in path directions for detecting neural periodicity. This strategy, referred to as “Radial Adjustment” involved making a one-unit adjustment to one feature with the left hand followed by a one-unit adjustment to the other feature with the right hand.

To examine participants’ movement strategy, we first calculated the number of unique directions across every three consecutive steps within their original movement paths (Fig. S1 [↗](#)). Participants showed an average of 3.8 directions per path, which was significantly higher than the two directions predicted under the “horizontal–vertical movement” strategy ($t(32) = 15.76$, $p < 0.001$, two-tailed; Cohen’s $d = 2.78$). Next, we examined the uniformity of directional distributions across all generated paths spanning 0°–360°. No significant deviation from uniformity was observed (Fig. S2 [↗](#); $p > 0.05$, Rayleigh test for circular uniformity). These results confirm that the “Radial Adjustment” strategy was stably adopted by participants during navigation in the conceptual Greeble space, ensuring that subsequent analyses were not biased by directional sampling.

To eliminate potential learning effect on potential BOLD signal periodicity, all participants completed a training task one day prior to the MRI scanning to familiarize themselves with Greebles morphing (see methods for details). Behavioral performance was defined as a composite score combining path length and error size, with superior performance characterized by smaller scores. Learning effects were observed during the practice experiment on day 1 (Fig. S3 [↗](#); $t(32) = -2.46$, $p = 0.019$, two-tailed; Cohen’s $d = 0.44$), but were no longer present during the MRI experiment on day 2 ($t(32) = -0.74$, $p = 0.462$, two-tailed; Cohen’s $d = 0.13$). These results suggest that participants maintained stable task accuracy across experimental sessions during MRI scanning. (Fig. 1d [↗](#); Paths of individual participants are shown in Fig. S4 [↗](#)).

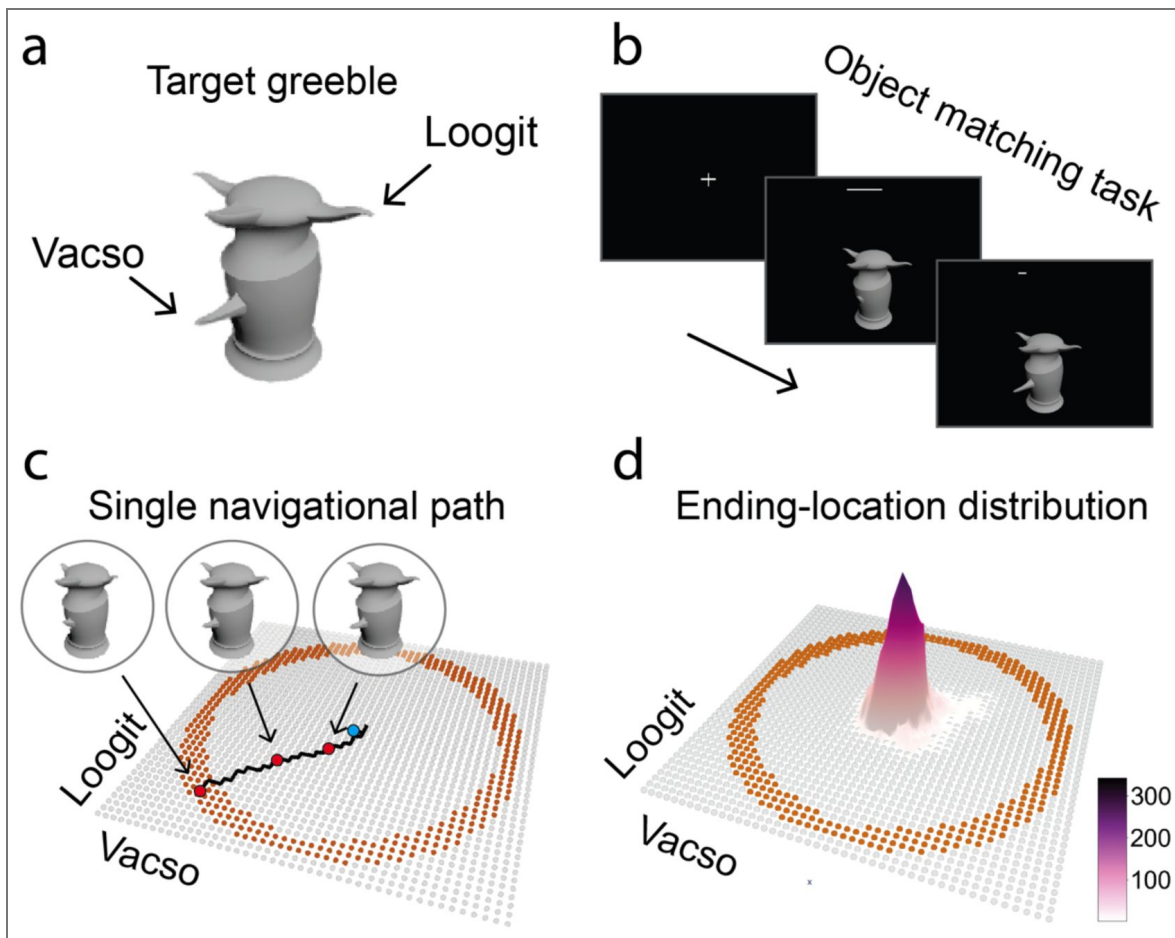


Fig. 1. Experimental design.

a) Depiction of the Greeble prototype (Gauthier and Tarr, 1997) and its two defining features, namely “Loogit” and “Vacso.” b) Inside the MRI scanner, participants adjusted the length of Loogit and Vacso to match the prototype by stepwise button presses, within a 10-s time limit. c) Conceptual object space. Each orange dot within the ring-shaped area represents a Greeble variant, while the central blue dot indicates the Greeble prototype (i.e., the goal location). The red dots denote exemplar intermediate locations along the navigational path (i.e., the black line). d) Density distribution of participants’ ending locations indicated an overall superior behavioral performance for detecting the periodic activity of the HPC.

6-fold periodicity in the EC

Using sinusoidal modulation, the grid orientation, reflecting the allocentric direction of the grid axes, was calculated for each voxel within the bilateral EC using half of the dataset (Experimental session 1, 3, 5, 7). Participants' path directions were determined by the 2D position of Greeble variants relative to the ending locations, with 0° arbitrarily defined as movement from the East (Fig. 2a). Significant deviations from uniformity were observed in 30 out of 33 participants (Fig. S5, $p < 0.05$; Rayleigh test of uniformity and pairwise phase consistency (PPC), Bonferroni-corrected across participants), indicating that voxelwise grid orientations in the EC were consistent for the majority of participants.

Hexagonal 6-fold activity was examined using the other half of the dataset (Experimental session 2, 4, 6, 8), with path directions calibrated according to participant-dependent grid orientations. This analysis identified a significant cluster in the right EC (Fig. 1b; initial threshold: $p = 0.05$, two-tailed; cluster-based SVC correction for multiple comparisons: $p < 0.05$; Cohen's $d = 0.63$; Peak MNI coordinate: 32, -6, -30). Four ROI-based control analyses were performed using rotationally symmetric parameters (3-, 4-, 5-, and 7-fold) to validate the robustness of the cluster in representing the 6-fold periodicity. The ROI was defined as a functional mask of the right EC identified in the voxel-based analysis and further restricted within the anatomical EC. These analyses revealed significant periodic modulation only at 6-fold (Fig. 2c; $t(32) = 3.56$, $p = 0.006$, two-tailed, corrected for multiple comparisons across rotational symmetries; Cohen's $d = 0.62$). In contrast, no significant effects were observed for 3-fold periodicity (120° periodicity; $t(32) = 1.10$, $p = 0.28$, Cohen's $d = 0.19$), 4-fold periodicity (90° periodicity; $t(32) = 0.31$, $p = 0.76$, Cohen's $d = 0.05$), 5-fold periodicity (72° periodicity; $t(32) = -0.21$, $p = 0.83$, Cohen's $d = -0.04$), or 7-fold periodicity (51.4° periodicity; $t(32) = 0.88$, $p = 0.39$, Cohen's $d = 0.16$).

We further examined the fluctuation of hexagonal activity across the angular domain of path directions, following the approach reported in previous studies (Doeller et al., 2010; Constantinescu et al., 2016; Bao et al., 2019; Wagner et al., 2023; Raithel et al., 2023). In these studies, EC activity exhibited six peaks as a function of path direction. To do so, we reconstructed the BOLD signals using sinusoidal modulation for each path direction and each participant. To account for individual difference in spatial phase, path directions were calibrated by subtracting the participant-specific grid orientation estimated from the odd sessions. The directional tuning curve was then derived by averaging the reconstructed BOLD signals across voxels within a hand-drawn bilateral EC mask and across participants. A stable 6-fold periodicity was observed (Fig. 2d, left; $p < 0.05$, permutation corrected for multiple comparisons), characterized by stronger activity when path directions were aligned with the calibrated grid axes and weaker activity when they were misaligned, confirming a 60° periodicity of EC activity across conceptual directions in the Greeble space (Fig. 2d, right).

3-fold periodicity in the HPC

Potential neural periodicity in the HPC was assessed using a Fast Fourier Transform (FFT)-based spectral analysis. FFT was employed to permit unbiased identification of multiple candidate periodicities (e.g., from 3- to 7-fold), independent of any a priori assumptions regarding spatial phase (orientation). Unlike the EC where a 6-fold periodicity is well established, the periodic structure of the HPC was not known a priori. FFT was thus applied as a phase-invariant approach to detect potential periodicities. Moreover, FFT was employed as an independent analysis to cross-validate the sinusoidal modulation results, thereby offering complementary evidence for the 6-fold periodicity in EC and the 3-fold periodicity in HPC. Accordingly, BOLD signals from all eight experimental sessions were modeled using a GLM to estimate direction-dependent activity. Participants' path directions were down-sampled into 10° bins, resulting in a total of 36 directional bins (e.g., 0°, 10°, 20°, etc.). These bins were entered into the GLM as binary regressors. The resulting direction-dependent parametric maps were sorted in ascending order of spatial directions, and FFT was applied to extract spectral magnitude maps for periodicities ranging from 3- to 7-fold (Fig. 3a; see Methods for details).

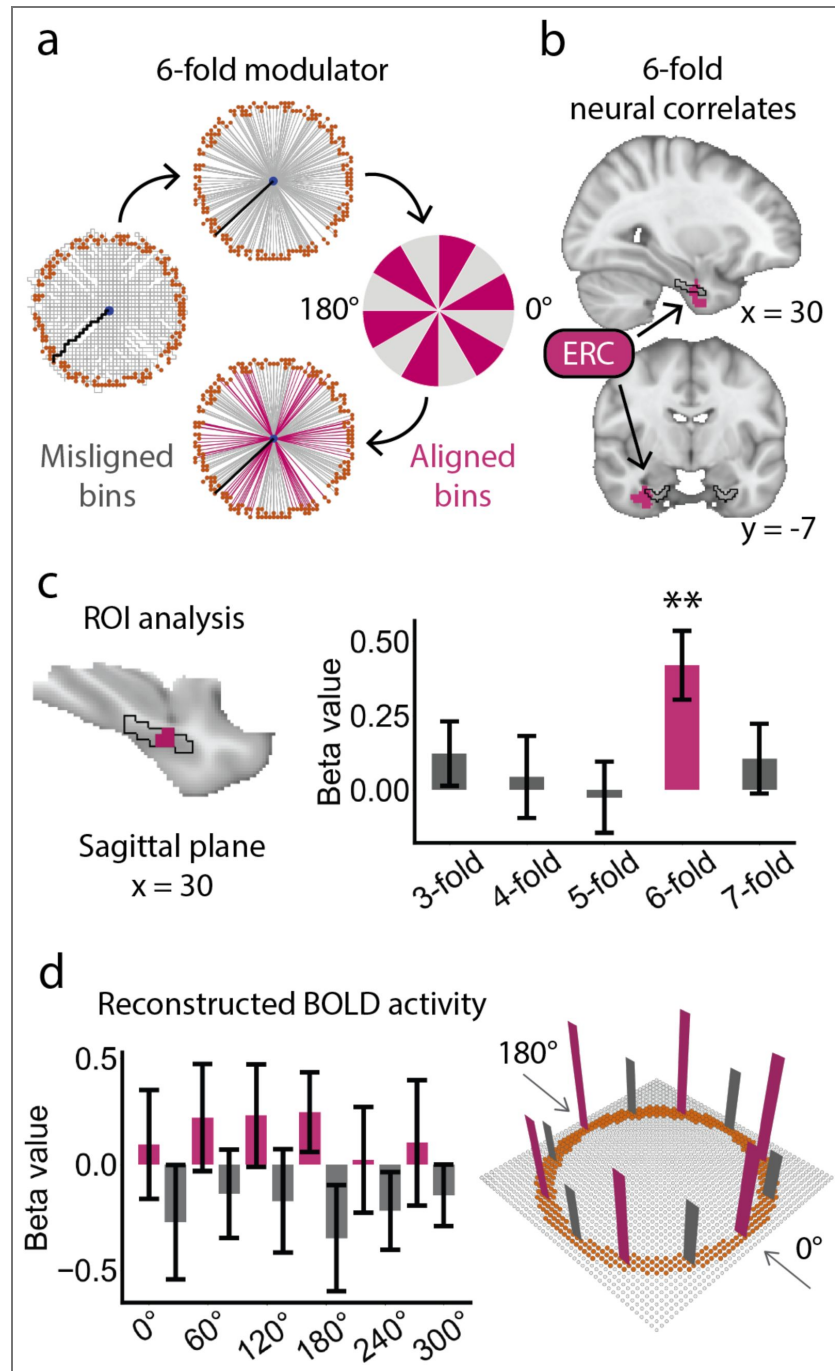


Fig. 2. 6-fold periodicity in the EC.

a) Schematic defining path directions in the 6-fold modulation. Participants' path directions (Top) were extracted from the original paths (Left) by connecting the starting and ending locations, with 0° arbitrarily set to the East as the reference, and were then classified as "aligned" or "misaligned" (Right & Bottom). The original paths of each participant were referred to Fig. S4. b) Voxel-based sinusoidal modulation revealed significant 6-fold periodicity within the right EC (Voxel-based analysis: initial threshold: $p = 0.05$, two-tailed; cluster-based small volume correction (SVC) for multiple comparisons: $p < 0.05$; Cohen's $d = 0.63$; Peak MNI coordinate: 32, -6, -30). Volumetric results are displayed in radiological orientation; numbers below the brain slices indicate MNI coordinates. c) ROI-based analysis, using a functional mask derived from the significant right EC cluster and constrained within the anatomical EC, confirmed a significant 6-fold periodicity ($t(32) = 3.56$, $p = 0.006$, two-tailed, corrected for multiple comparisons across rotational symmetries; Cohen's $d = 0.62$). The black line indicates the boundary of the EC. d) Schematic illustrating the 6-fold directional tuning curve reconstructed from sinusoidal modulation (Left) and its representation in the 2D Greeble space (Right).

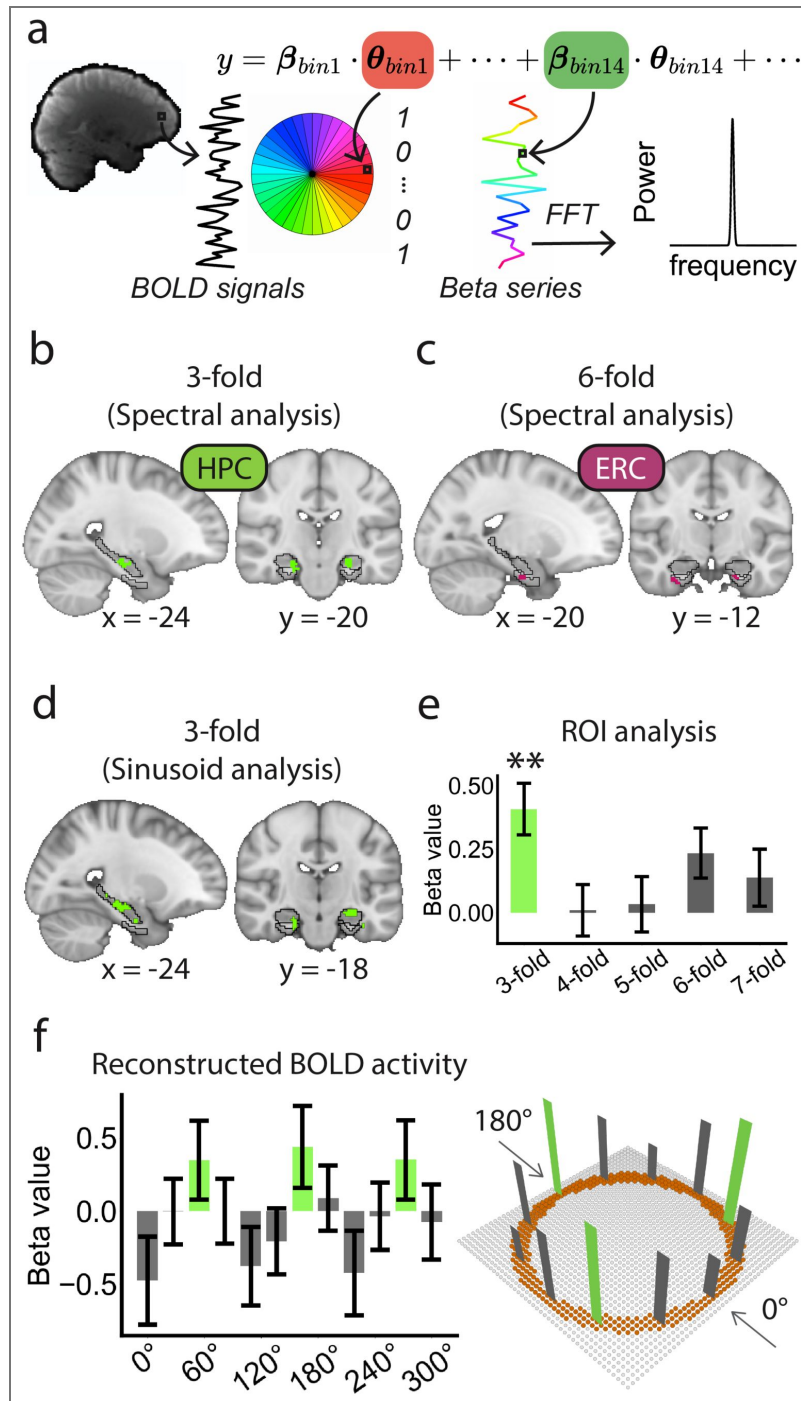


Fig. 3. 3-fold periodicity in the HPC.

a) Schematic illustration of the spectral analysis procedure. b-c) Voxel-based spectral analysis revealed significant 3-fold periodicity in the bilateral HPC and 6-fold periodicity in the bilateral EC (initial threshold: $p = 0.05$, two-tailed; Cluster-based SVC correction for multiple comparisons: $p < 0.05$; For the HPC: Cohen's $d = 1.06$; Peak MNI coordinate: -24, -20, -18; For the EC: Cohen's $d = 1.27$; Peak MNI coordinate: -22, -14, -30). The black lines indicate the boundaries of the HPC and EC. d-e) 3-fold periodicity in the bilateral HPC identified using sinusoidal modulation (initial threshold: $p = 0.05$, two-tailed; cluster-based SVC correction: $p < 0.05$; Cohen's $d = 0.68$; peak MNI coordinate: -24, -18, -12). ROI-based analysis using a functional mask of this cluster within the anatomical HPC confirmed the effect ($t(32) = 3.94$, $p = 0.002$; Cohen's $d = 0.70$). Volumetric results are displayed in radiological orientation; numbers below the brain slices indicate MNI coordinates. f) Schematic illustrating the 3-fold directional tuning curve reconstructed from sinusoidal modulation (Left) and its representation in the 2D Greeble space (Right).

A significant cluster of 3-fold periodicity was observed in the bilateral HPC (Fig. 3b; initial threshold: $p = 0.05$, two-tailed; Cluster-based SVC corrected for multiple comparisons: $p < 0.05$; Cohens' $d = 1.06$; Peak MNI coordinate: -24, -20, -18). Moreover, a significant cluster of 6-fold periodicity was identified in the right EC (Fig. 3c; initial threshold: $p = 0.05$, two-tailed; Cluster-based SVC-corrected for multiple comparisons: $p < 0.05$; Cohen's $d = 1.27$; Peak MNI coordinate: -22, -14, -30), thereby replicating the finding from the sinusoidal modulation. No significant periodic activity was observed for the other folds ($P > 0.05$).

The hippocampal 3-fold periodicity has not been previously reported in the spatial domain by either neurophysiological or fMRI studies. To validate its reliability, we conducted additional analyses. First, sinusoidal modulation confirmed significant 3-fold clusters in the bilateral HPC (Fig. 3d; initial threshold: $p = 0.05$, two-tailed; Cluster-based SVC-corrected for multiple comparisons: $p < 0.05$; Cohen's $d = 0.68$; Peak MNI coordinate: -24, -18, -12). ROI analysis, using a functional mask of the HPC identified in the spectral analysis and further restricted within the anatomical HPC, indicated that HPC activity selectively fluctuated at 3-fold periodicity (Fig. 3e; $t(32) = 3.94$, $p = 0.002$, corrected for multiple comparisons across rotational symmetries; Cohen's $d = 0.70$), whereas no significant activity periodicity was observed at other spatial fold (4-fold periodicity: $t(32) = 0.09$, $p = 0.93$, Cohen's $d = 0.02$; 5-fold periodicity: $t(32) = 0.31$, $p = 0.76$, Cohen's $d = 0.05$; 6-fold periodicity: $t(32) = 2.36$, $p = 0.12$, Cohen's $d = 0.42$; 7-fold periodicity $t(32) = 1.21$, $p = 0.24$, Cohen's $d = 0.21$). The reconstructed BOLD signals, direction-calibrated for each participant based on their spatial phase estimated from sinusoidal modulation, showed a significant 3-fold periodic pattern in the bilateral HPC (Fig. 3f, left; $p < 0.05$, corrected for multiple comparisons). In sum, both the spectral and sinusoidal modulation confirmed a 3-fold periodic activity in the HPC across the conceptual directions within the 2D Greeble space (Fig. 3f, right), which was distinct from the 6-fold periodicity observed in the EC.

Second, the 3-fold HPC periodicity was not affected by the precision of directional sampling, as consistent findings were observed when 20° bins were applied (Fig. S6; initial threshold: $p = 0.05$, two-tailed; Cluster-based SVC corrected for multiple comparisons: $p < 0.05$; Cohen's $d = 1.18$; Peak MNI coordinate: 22, -24, -14). Third, we independently analyzed HPC and EC periodicity using spectral analysis for each of the three site-dependent experimental groups (See methods for details). These analyses showed reliable activity periodicities in both brain areas across experimental groups (Fig. S7; initial threshold: $p = 0.05$, two-tailed. Cluster-based SVC corrected for multiple comparisons: $p < 0.05$). Fourth, we further examined the whole brain representation of both 3- and 6-fold activity periodicity, respectively (Fig. S8). The 3-fold periodicity revealed significant involvement of the medial prefrontal cortex (mPFC), precuneus (PCu), and parietal cortex (PC) (initial threshold: $p = 0.05$, two-tailed. Whole brain correction for multiple comparisons: $p < 0.05$), suggesting engagement of the default mode network (DMN). In contrast, the 6-fold periodicity highlighted the Salience network, including the anterior cingulate cortex (ACC) and insular cortex (INS), in addition to the EC (initial threshold: $p = 0.05$, two-tailed; Whole brain cluster-based correction for multiple comparisons: $p < 0.05$). These results demonstrated distinct functional networks involved in integrating spatial metric (6-fold periodicity) and in judging self-motion within conceptual space (3-fold periodicity).

Phase synchronization between the HPC and EC activity

To examine whether the spatial phase structure in one region could predict that in another, we tested whether the orientations of the 6-fold EC and 3-fold HPC periodic activities, estimated from odd-numbered sessions using sinusoidal modulation with rotationally symmetric parameters (in radians), were correlated across participants. A cross-participant circular-circular correlation was conducted between the spatial phases of the two areas to quantify the spatial correspondence of their activity patterns (EC: purple dots; HPC: green dots) (Jammalamadaka & Sengupta, 2001). The analysis revealed a significant circular correlation (Fig. 4a; $r = 0.42$, $p < 0.001$), as reflected by the continuous color progression across the participants (i.e., the colored lines connecting each

pair of the EC and HPC dots in Fig. 4a [↗](#)), suggesting that participants with smaller hippocampal phases (green, outer ring) tended to have smaller entorhinal phases (purple, inner ring), and vice versa.

In addition to the across-participant phase correlation, we further examined the spatial alignment between the 6-fold EC and 3-fold HPC activity patterns. Given that the spatial phase of the HPC depends on EC's projections as we hypothesized, particularly along the three primary axes of the hexagonal codes, we examined whether the periodic activities of the EC and HPC were peak-overlapped? Notably, unlike previous studies that examined temporal coherence of neural oscillations (Buzsaki, 2006 [↗](#); Maris et al., 2011 [↗](#); Fries et al., 2013 [↗](#)), our analysis focused on the periodic coupling of brain areas in spatial direction domain. To test this, a cross-frequency spatial coupling analysis (adapted from the amplitude–phase coupling framework; Canolty et al., 2006 [↗](#)) was employed to identify at which HPC phase the EC exhibited maximal amplitude modulation, quantified by a modulation index M (Canolty et al., 2006 [↗](#)). If the activities of both areas were peak-overlapped (i.e., no peak offset), a strong coupling at phase 0 of the HPC would be expected as shown by the one-cycle-based schema in Fig. 4b [↗](#).

The BOLD signals of both regions were first reconstructed using sinusoidal modulation based on participants' path directions, without calibrating their spatial phase estimated from rotationally symmetric parameters. The 6-fold EC and 3-fold HPC activity were then converted into analytic representations using the Hilbert transform to extract their instantaneous phase of the HPC (Φ_{HPC}) and amplitude envelope of the EC (A_{ERC}). The phases of the HPC were classified into nine bins. The composite analytic signal, defined as $z = A_{ERC}e^{i\Phi_{HPC}}$, was then used to compute the modulation index M , defined as the absolute value of the mean of z values within each bin, quantifying the scalar coupling strength between EC amplitude and HPC phase. To assess statistical significance, a surrogate dataset, a null distribution of the modulation indices (M'), was generated by spatially offsetting the EC amplitude relative to the HPC phase across all possible spatial lags. This procedure preserved the within-region structure while disrupting inter-regional alignment. The mean of this surrogate distribution was used as the baseline reference against which the observed coupling strength was compared. As the result, significant coupling was observed in the bin centered at phase 0 of the HPC (Fig. 4c [↗](#); $t(32) = 2.57$, $p = 0.02$, Bonferroni-corrected across tests; Cohen's $d = 0.45$). In contrast, no significant coupling was found in other bins ($p > 0.05$). To rule out the possibility that the observed coupling was driven by the integer multiple (harmonic) relationship between the 3-fold and 6-fold periodicities, we additionally conducted control analyses using 9-fold and 12-fold EC components. However, no significant coupling was observed in these controls (Fig. 4c [↗](#); $P > 0.05$). Together, these results confirmed selective alignments of spatial peaks between the 6-fold EC and 3-fold HPC periodicity in the conceptual direction domain.

3-fold periodicity in human behavior

Considering the reciprocal connectivity between the medial temporal lobe (MTL), where the EC and HPC reside, and the parietal cortex implicated in visuospatial perception and action, together with the observed 3-fold periodicity within the DMN (including the PC and PCu; Fig. S8 [↗](#)), we hypothesized that the 3-fold periodic representations of path directions extend beyond the MTL to the egocentric cortical areas, such as the PC, thereby influencing participants' visuospatial task performance (Fig. 5a [↗](#)). To test this hypothesis, we assessed participants' behavioral performance using two metrics: (1) path length and (2) the deviation (i.e., error size) between the actual ending location and the goal location per trial, since participants rarely stopped exactly at the goal. This procedure yielded, for each participant, a performance vector in the directional domain (Fig. 5b [↗](#)), with superior performance indexed by a composite score reflecting shorter path and smaller error size.

Using spectral analysis, we identified a significant periodicity in participants' behavioral performance, with spectral power peaking at a 3-fold symmetry (Fig. 5c [↗](#); $p < 0.05$, permutation corrected for multiple comparisons), demonstrating that participants' visuospatial perception performance fluctuated as a function of path directions. In contrast, no significant spectral power was observed at other spatial folds, including 6-fold behavioral periodicity ($p > 0.05$). As a control,

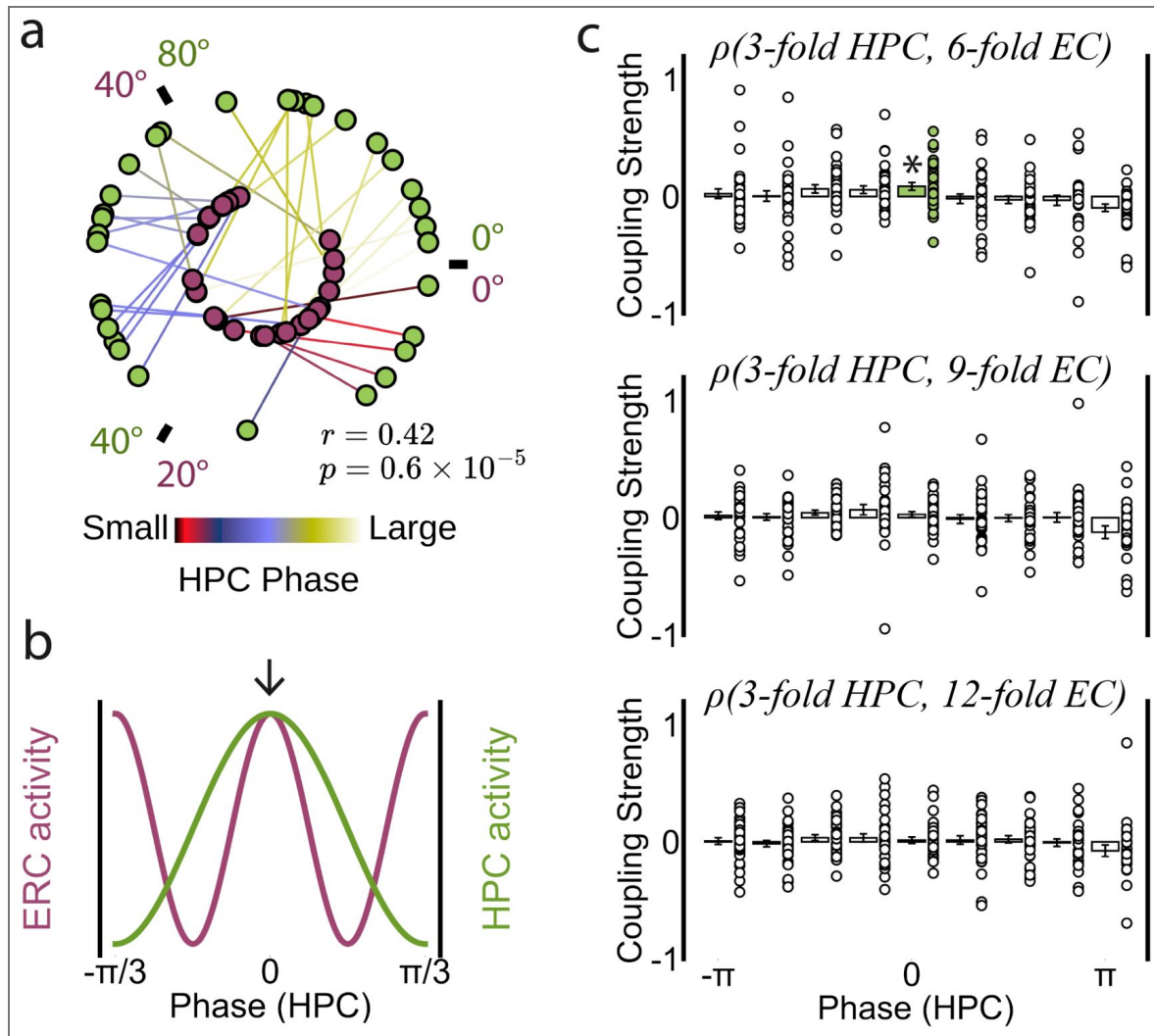


Fig. 4. Phase synchronization between the HPC and EC activity in the directional domain.

a) Cross-participant circular-circular correlation analysis revealed a significant coupling between the EC and HPC phases ($r = 0.42$, $p < 0.001$). The green (outer ring) and purple (inner ring) dots denote HPC and EC phases, respectively. Each line connecting the EC and HPC phases represents one participant, with its color indicating the HPC phase (0 – 120°). b) Schematic illustration of a hypothetical peak-overlapping pattern (blue ellipse) between the HPC and EC activity in corresponding to spatial phase (one-cycle). c) Amplitude-phase modulation analysis revealed significant coupling between the 3-fold HPC activity and 6-fold EC activity in the bin centered at phase 0 ($t(32) = 2.57$, $p = 0.02$, Bonferroni-corrected across tests; Cohen's $d = 0.45$). The coupling strength was computed as the difference between the observed modulation index (M) and the mean surrogate modulation index (M'). No significant coupling was observed in the control analyses testing phase synchronization between the 3-fold HPC and the 9-fold or 12-fold EC periodicities ($p > 0.05$). The error bars indicate SEM. ***: $p < 0.001$.



Fig. 5. 3-fold periodicity in behavioral performance.

a) Schematic illustration of visuospatial task performance potentially inheriting HPC's 3-fold periodicity. b) Participants' behavioral performance, measured by a composite index of path length and deviation from the goal to ending locations, fluctuated as a function of path directions. The shaded area denotes SE. c) Spectral analysis revealed significant power at the 3-fold of participants' behavioral performance ($p < 0.05$, corrected for multiple comparisons). The red dashed line represents the baseline derived from permutation. d) Significantly higher phase-locking values were observed between participants' behavioral performance and HPC activity compared to surrogate dataset ($t(32) = 8.10$, $p < 0.001$; Cohen's $d = 1.14$).

we also examined behavioral periodicity along each individual metric, path length and error size. A significant 3-fold periodicity was observed for path length (Fig. S9 [↗](#); $p < 0.05$, permutation-corrected for multiple comparisons), whereas no significant modulation was found for error size, likely reflecting a residual 4-fold bias inherent to the four cardinal-direction movement structure. To further investigate the causal relationship in brain-behavior coupling, we calculated the phase-lag index between the 3-fold behavioral periodicity and HPC activity for each participant (Stam et al., 2007 [↗](#)). This analysis revealed a significantly higher phase-locking value (PLV) compared to surrogate dataset (Fig. 5d [↗](#); $t(32) = 8.10$, $p < 0.001$; Cohen's $d = 1.14$), confirming phase coupling between the 3-fold HPC activity and the 3-fold behavioral performance. Contrary to the commonly held assumption of isotropic task performance across path directions, these results demonstrated a biased behavioral pattern influenced by HPC and EC activity periodicity, with superior navigational performance (i.e., shorter path length and smaller error size) observed when path directions aligned with grid axes compared to when they were misaligned.

The “EC-HPC PhaseSync” Model

The above empirical results elucidate that the vectorial representations of the HPC, characterized by 3-fold activity periodicity, were spatially coherent with EC population activity across the directional domain. To mechanistically explain how the 6-fold EC code gives rise to a 3-fold HPC representation, we constructed a cognitive model, termed the “EC-HPC PhaseSync” model (Fig. 6 [↗](#)), which demonstrates the possible computational mechanism of the EC-HPC circuit, through simulating how the periodic representations emerged in the HPC influences spatial movements toward the center of Greeble space, as well as qualifying the reliability of the 3-fold periodic performance, particularly when goal locations are randomly distributed across the conceptual space.

The EC grid code G was simulated using a cosine grating model (O’keefe and Burgess, 2005 [↗](#); Burgess et al., 2007 [↗](#); Blair et al., 2007 [↗](#); Bush and Schmidt-Hieber, 2018 [↗](#); see methods for details). A population of 45-by-45 grid cells were generated to tessellate the space (Fig. 6a [↗](#), left), with spatial orientations kept constant while spatial phases varied across locations. The population activity in the EC, associated with movements from a start location, was termed the path code V , simulated by summing the grid codes across locations along the path (Fig. 6a [↗](#), right). Notably, the path code V is exactly identical for path direction Φ and $\Phi + 180^\circ$, naturally reflect the spatial orientation $\psi \in [0, \pi)$ across the space.

The downstream HPC activity was simulated using a two-dimensional δ vector to represent spatial orientation. Each δ value, encoded by a single HPC neuron, was obtained by linearly summing EC activity across locations along the path code V (Fig. 6b [↗](#), black dots). The spatial periodicity and spatial phase of δ depend on the number of primary axes and orientation of grid codes, while the magnitude of δ indicates the degree of alignment between the path orientation ψ and the grid axes. Therefore, the 3-fold structure of δ , projected from the EC grid cells, initially represents environmental orientations when self-location is not integrated. When self-location is anchored into space, this 3-fold δ structure embeds allocentric directions so that the preferred directions can be distinguished. The amplitude of δ reflects the directional modulation strength and the phase specifies the angular offset of the 3-fold pattern. The goal-directed vector representation C in the HPC was next simulated by integrating the δ vector with a Gaussian-based distance vector centered at the goal location, thereby driving movements toward the goal (Fig. 6c [↗](#)).

During simulations, 100 goal locations were randomly generated within the space, each paired with 120 starting locations, arranged in a ring-shaped pattern near the boundary. At each simulation step, a vector of HPC activity was extracted from the locations surrounding the current location, and the next movement was determined by the strongest activity of adjacent locations, following winner-take-all dynamics. The step size, initialized as 1 pixel, was dynamically adjusted using a stepwise approach, with a maximum limit of 5 pixels; for example, the step size increased by 1 pixel if no stronger activity was found among the surrounding locations compared to the activity of self-location. Model performance was defined by the path length, calculated as the sum of Euclidean distances between consecutive locations, with superior performance corresponding

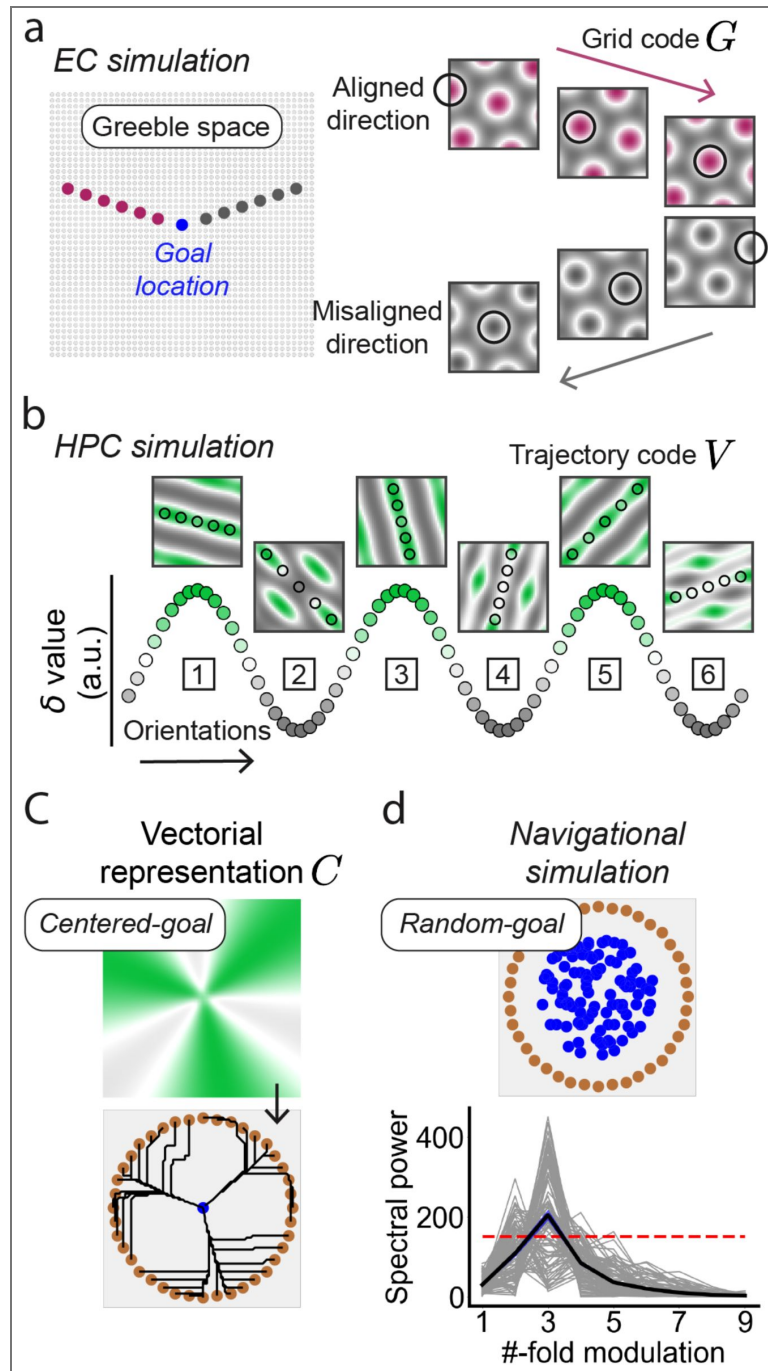


Fig. 6. The EC-HPC PhaseSync Model.

a) Schematic illustrating the population activity of grid cells during mental planning. The simulated grid cell population was activated by visiting discrete locations (the black circles in the right panel) in the Greeble space along directions either aligned (purple) or misaligned (grey) relative to the grid axes. b) The 3-fold periodicity of the path code V represented in the HPC. A path code V is symmetrical for path direction Φ and $+180^\circ$, representing a unique spatial orientation ψ ranging from 0° to 180° . The δ value (y-axis) indicates the degree to which the spatial orientation ψ of a path code V aligns with the grid axes, with larger δ values (e.g., the spatial orientation (1, 3, and 5)) indicating a perfect match between Φ and grid axes. c) Simulated vectorial representation of the HPC for centered-goal-based navigation. The 3-fold periodicity is driven by vectorial gradients inherited from δ . d) Random-goal-based navigational simulation. Significant spectral power of model performance was observed at 3-fold across 100 randomly selected goal locations (the blue dots and gray lines) ($P < 0.05$, corrected for multiple comparisons). The red dashed line indicates the significance threshold derived from permutation. The blue shaded areas denote the standard error. Grey lines represent the spectral powers of goal-dependent simulations.

to shorter path length. The model performance of all directions was sorted in ascending order before spectral analysis, and the mean spectral powers, calculated across 100 navigation tasks, was compared with a permutation-based significant threshold. The simulations revealed a significant 3-fold periodicity in path length across 100 navigation tasks (Fig. 6d; $P < 0.05$, corrected for multiple comparisons). No other periodicities reached significance ($P > 0.05$). These results demonstrate that projections from grid cell populations can reproduce the 3-fold behavioral periodicity observed in humans during goal-directed navigation.

Discussion

Spatial navigation, the cognitive process of retrieving spatial relationships from one moment to the next in both conceptual (Epstein et al., 2017; Behrens et al., 2018; Bottini and Doeller, 2020; Park et al., 2020) and physical spaces (Tolman, 1948; Sargolini et al., 2006; Gardner et al., 2022; Ormond and O’Keefe, 2022), relies on the EC-HPC circuit. These two areas are known to have distinct functions. The EC encodes continuous space with a metric representation (Hafting et al., 2005), while the HPC represents localized spatial locations (O’Keefe and Dostrovsky, 1971). A longstanding puzzle is how they collaborate to encode spatial navigation. In the present study, we modulated the EC and HPC activity using conceptual directions, and identified 6-fold periodicity from EC activity and 3-fold periodicity from HPC activity. The periodic activities of both areas were synchronized and peaked-overlapped. Additionally, 3-fold periodicity was also observed in participants’ behavioral performance, which was synchronized with the HPC activity, suggesting a direct influence of periodic brain activity on visuospatial perception. Our findings provide a potential explanation for the observation of the hippocampal vector field (Sarel et al., 2017; Ormond and O’Keefe, 2022; Muhle-Karbe et al., 2023).

In addition to the HPC and EC, whole-brain spectral analysis revealed involvement of the DMN and salience network for the 3- and 6-fold periodicity, respectively, suggesting that concept-dependent periodic neural activity is supported by distinct networks and extends beyond the EC-HPC circuit. These results do not imply that grid cell’s hexagonal code is the origin of the spatially selective periodic activity observed in these brain networks. Rather, these large-scale activity dynamics more likely reflect hierarchical computations across the brain. In upstream visual areas, neurons encode position-specific features and provide egocentric cues when an allocentric map is unavailable, as seen in the feature-selective neurons in V1 that represent locations across visual space, as described by Hubel and Wiesel (1959, 1962). Moreover, spatial attention has been shown to operate through rhythmic sampling in the temporal domain. Neural activity in the frontoparietal attention networks oscillates at 3–8 Hz and predicts trial-by-trial performance in spatial discrimination tasks (Fiebelkorn et al., 2018). This finding parallels our observed 3-fold behavioral periodicity, indexed by task performance, suggesting that neural representation in both the temporal and spatial domains may rely on periodic computations, potentially reflecting a shared mechanism by which the brain organizes perception and guides goal-directed behavior.

The EC-HPC PhaseSync model demonstrates how a vectorial representation may emerge in the HPC from the projections of populations of periodic grid codes in the EC. The model was motivated by two observations. First, the EC intrinsically serves as the major source of hippocampal input (Witter and Amaral, 1991; van Groen et al., 2003; Garcia and Buffalo, 2020), and grid codes exhibit nearly invariant spatial orientations (Hafting et al., 2005; Gardner et al., 2022). Second, mental planning, characterized by “forward replay” (Dragoi and Tonegawa, 2011; Pfeiffer, 2020), has the capacity to activate populations of grid cells that represent sequential experiences in the absence of actual physical movement (Nyberg et al., 2022). We hypothesize that an integrated path code of sequential experiences may eventually be generated in the HPC, providing a vectorial gradient toward the goal location. The path code exhibits regular, vector-like representations when the path direction aligns with the orientations of grid axes, and becomes irregular when they misalign. This explanation is consistent with the band-like representations observed in the dorsomedial EC (Krupic et al., 2012) and the irregular activity fields of trace cells in the HPC (Poulter et al., 2021).

In sum, our study identifies a 3-fold hippocampal code that complements the 6-fold EC metrics, potentially supporting flexible navigation. Extending previous cognitive models based on grid- and place-cell simulations of navigation (Bicanski and Burgess, 2019; Whittington et al., 2020; Edvardson et al., 2020), our findings highlight a potential hierarchical periodic architecture of neurons within the MTL, particularly in representing spatial directions. Moreover, considering that the task space was simplified to an idealized square layout. In more complex, real-world environments, hippocampal three-fold periodicity may interact with other spatial variables such as distance, movement speed, and environmental boundaries. Future studies should aim to characterize how the periodic activity of the EC-HPC circuit interacts with the external environment to support real-world navigation.

Methods

Participants

Thirty-three right-handed university students with normal or corrected-to-normal vision participated in the experiment. Participants were recruited in three separate batches: 10 from Tsinghua University (Group 1: mean age = 23.9 [SD = 3.51], 5 females, 5 males), 10 from Peking University (Group 2: mean age = 22.10 [SD = 2.23], 5 females, 5 males), and 13 from Tsinghua University (Group 3: mean age = 20.85 [SD = 3.08], 2 females, 11 males). All participants had no history of psychiatric or neurological disorders and gave their written informed consent prior to the experiment, which received approval from the Research Ethics Committee of both universities.

Greeble space

The Greebles (Gauthier and Tarr, 1997) were generated based on the modified codes of the TarrLab stimulus datasets (<http://www.tarrlab.org/>) using Autodesk 3ds Max (version 2022, <http://www.autodesk.com>). A Greeble's identity was defined by the lengths of two distinct features, represented by two nonsense words, "Loogit" and "Vacso" (Fig. 1a). These nonsense names were arbitrarily assigned and carried no semantic relevance to the task. The generated Greebles were arranged in a 45-by-45 matrix, forming a two-dimensional space, with each Greeble representing a "location". The feature 'Loogit' corresponded to the y-axis, while the feature 'Vacso' represented the x-axis (Fig. 1c). The prototype Greeble, positioned at the center of the Greeble space (Fig. 1c; blue dot), served as the target (i.e., "goal location"), with Loogit and Vacso having equal lengths. The length ratios of Greebles' feature relative to the Greeble prototype ranged from 0.12 to 1.88, with a step size of 0.04, providing a smooth morphing experience as participants adjusted the lengths of the Greeble features.

Experimental design

The object matching task was programmed using Pygame (version 2.0, <https://www.pygame.org/>; Python version 3.9). Each trial began with a 2.0-s fixation screen, followed by a pseudo-randomly selected Greeble variant presented at the center of the screen (Fig. 1b). These variants were sampled from those near the boundary of the feature space (Fig. 1c; orange dots). Participants were instructed to adjust the features of the Greeble variant by pressing buttons to match the Greeble prototype as quickly as possible. Two response boxes enabled participants to stretch or shrink the two features. These boxes were counterbalanced between the left and right hands of participants. To mitigate the impact of learning effects on BOLD signals, participants completed a training task one day prior to MRI scanning. On the training day, participants completed the task in a self-paced manner. They first familiarized themselves with the task and then performed at least 10 practice trials before the main training session, freely adjusting Greeble features and pressing the return key to end each trial. Feedback was provided during the first 7 sessions (e.g., "Loogit: 5 percent longer than the target"), while no feedback was given in the final session. Day 2's MRI experiment procedure mirrored day 1, except that 1) participants were required to complete a trial within 10 seconds. A timer was presented on top of screen (Fig. 1b), and 2) no feedback was provided for all experimental sessions. Based on the present design, the "path" from

a Greeble variant towards the prototype can be interpreted as forming a conceptual direction. To ensure an even distribution of conceptual directions, we arbitrarily defined the “movement” from the East as 0°. For each 30° bin, 24 Greeble variants were pseudo-randomly selected from the boundary of the feature space with an angular precision of 1.25°, resulting in a total of 288 Greeble variants distributed in 8 experimental sessions. Each session contained 36 object matching trials and 4 lure trials. The lure trial presented a blank screen after the fixation for 10 seconds. The MRI experiment lasted approximately 1 hour. Participants were incentivized to perform accurately with exponentially increasing monetary rewards based on their performance. After the scan, participants were debriefed, shown their performance results, and asked to discuss their subjective feeling as well as any perceived reasons for any performance issues. Importantly, participants were unaware of the existence of the Greeble space or the spatial features such as the location and direction during the task.

Behavioral performance

The behavioral performance score was calculated for each trial as $T - T' + E/0.04$, where T and T' denote the participants' actual and the objectively optimal path length, respectively. The term $E/0.04$ quantified the error size, the Euclidean distance (in pixels) between the participant's ending locations (final adjusted Greeble) and the goal location (Greeble prototype). The constant “0.04” represents the step size of the Greeble feature ratio, and was used to convert the ratio-based error to pixel units. The actual path length T was defined as the number of movement steps (in pixels) recorded from the participant's paths, whereas the optimal path length T' was determined by the number of pixels in the shortest possible path between the Greeble variants and Greeble prototype. Subtracting T' from T thus normalized each participant's performance relative to the optimal path. The rationale for this correction was to eliminate the systematic difference in path length between cardinal (i.e., “East”, “South”, “West”, and “North”) and intercardinal (e.g., “Southeast”) directions, which arises from the restricted movements along the horizontal and vertical axis of the squared Greeble space. In this design, movements along cardinal directions inherently result in shorter paths, while those along intercardinal directions become longer. Although this normalization effectively reduced the directional bias in path length, it did not remove the residual 4-fold bias in error size. Nevertheless, because path length and error size capture complementary aspects of task performance—navigational efficiency and positional accuracy—we incorporated both terms in quantifying behavioral performance. Superior performance was represented by shorter paths and smaller errors, corresponding to lower performance scores.

fMRI data acquisition

Imaging data were collected using a 3T Siemens Prisma scanner, equipped with a 64-and 20-channel receiver head coil at Tsinghua University and Peking University Imaging Center, respectively. Functional data were acquired with a Multi-band Echo Planar imaging (EPI) sequence with the following parameters: Acceleration Factor: 2, TR = 2000 ms, TE = 30 ms, matrix size = 112 × 112 × 62, flip angle = 90°, resolution = 2 × 2 × 2.3 mm³, 62 slices with a slice thickness of 2 mm and a gap of 0.3 mm, in a transversal slice orientation. Additionally, high-resolution T1-weighted three-dimensional anatomical datasets were collected for registration purposes using MPRAGE sequences, detailed as follows: TR = 2530 ms, TE = 2.98 ms, matrix size = 448 × 512 × 192, flip angle = 7°, resolution = 0.5 × 0.5 × 1 mm³, 192 slices with a slice thickness of 1 mm, in the sagittal slice orientation. At Tsinghua University, stimuli were presented through a Sinorad LCD projector (Shenzhen Sinorad Medical Electronics) onto a 33-inch rear-projection screen. At Peking University, stimuli presentation was managed using a visual/audio stimulation system (Shenzhen Sinorad SA-9939) and an MRI-compatible 40-inch LED liquid crystal display, custom-designed by Shenzhen Sinorad Medical Electronics Co., Ltd. The screen resolution for both sites was 1024 × 768. Participants viewed the stimuli through an angled mirror mounted on the head coil.

fMRI data preprocessing

The BOLD signal series of each scanning session was preprocessed independently using the FSL FEAT toolbox from the FMRIB's Software Library (version 6.0, <https://fsl.fmrib.ox.ac.uk/fsl/fslwiki>) (Smith et al., 2004; Woolrich et al., 2009; Jenkinson et al., 2012). For each scanning sessions, the BOLD signals were corrected for motion artifacts, slice time acquisition differences, geometrical distortion using fieldmaps, and were applied with a high-pass filter with a cutoff of 100 seconds. Spatial smoothing was performed using a Gaussian kernel with a full width at half maximum (FWHM) of 5 mm. For group-level analysis, the preprocessed BOLD signals were registered to each participant's high-resolution anatomical image and subsequently normalized to the standard MNI152 image using FSL FLIRT (Jenkinson and Smith, 2001). During normalization, the functional voxels were resampled to the resolution of 2 x 2 x 2 mm.

Sinusoidal modulation

A GLM with sinusoidal regressors was employed to investigate the periodic neural activity in the EC and HPC, following previously established protocols (Doeller et al., 2010; Constantinescu et al., 2016; Bao et al., 2019; Wagner et al., 2023; Raithel et al., 2023). First, the grid orientation in each voxel within the EC from each participant was calculated using half of the dataset (odd-numbered sessions: 1, 3, 5, 7). Specifically, a GLM was created for each session to model the BOLD signals of the EC with two parametric modulators: $\sin(6\theta)$ and $\cos(6\theta)$, convolved with the Double-Gamma hemodynamic response function. θ represents the path directions, derived from drawing a line between the starting and the ending location, as each raw adjustment step was limited to four cardinal directions. Defining direction by start–end vectors thus allowed full 0–360° coverage, consistent with previous grid-like coding studies (e.g., Constantinescu et al., 2016). The factor “6” represents the rotationally symmetric 6-fold neural activity. The factors “3”, “4”, “5” and “7”, misaligned with the primary axes of grid cells, were used as control parameters. The estimated weights $\beta_{\sin}$ and $\beta_{\cos}$ were then used to compute the grid orientation ϕ , ranging from 0° to 59°, where $\phi = [\arctan(\frac{\beta_{\sin}}{\beta_{\cos}})]/6$. The ϕ maps of each participant were averaged across the voxels of a hand-drawn bilateral EC mask and odd-numbered sessions to represent the grid orientation. Second, the neural representation of hexagonal activity, modulated by path directions in Greeble space, was examined by another GLM using the other half of the dataset (even-numbered sessions: 2, 4, 6, 8), with path direction calibrated by the participant-specific grid orientation. The GLM was specified with a cosine parametric modulator, $\cos(6[\theta - \Phi])$. Third, the hexagonal activity maps were averaged across the even-numbered sessions for each participant before entering second-level analysis. The same procedure was applied to the bilateral HPC to test for 3-fold periodicity. Specifically, voxel-wise phase estimates were obtained from odd sessions using $\sin(3\theta)$ and $\cos(3\theta)$ regressors, and the participant-specific 3-fold phase was obtained by averaging voxel-wise estimates across the bilateral HPC, using a mask derived from the AAL template. In the even sessions, path directions were rotationally calibrated by subtracting this participant-specific phase, and a GLM with $\cos(3[\theta - \Phi])$ was fitted. The resulting maps of 3-fold periodic modulation were averaged across even sessions for each participant and then submitted to group-level analysis. To present the directional tuning curves of EC and HPC activity in the angular domain of path directions, we reconstructed the BOLD signals for each participant using parameters estimated from sinusoidal modulation. Functional masks derived from significant subregions identified in the voxel-wise analyses were employed to capture their tuning profiles with maximum sensitivity.

Spectral analysis of MRI BOLD signals

Spectral analysis was employed to examine the spatial periodicity of BOLD signals. This procedure consisted of three stages: First-level analysis. A GLM was constructed to estimate direction-dependent activity maps. Participants' path directions were down-sampled into 36 bins of 10° each (e.g., path directions between 0° and 10° were assigned to the 0° bin, whereas those between 350°

and 360° were assigned to the 350° bin). These bins were entered as regressors in the GLM, yielding activity estimates associated with each directional bin for every participant. Second-level analysis. The direction-dependent activity maps were sorted in ascending order (from 0° to 360°), detrended, and processed with a Hanning window. They were then transformed from the spatial to the frequency domain using FFT (implemented in the stats package of R, version 4.0; <https://www.r-project.org/>) (Landau and Fries, 2012). The absolute values of the complex FFT outputs were used to generate spectral magnitude maps for each participant, covering periodicities from 0- to 18-fold. Third-level analysis. For group inference, the 95th percentile (two-tailed) of the pseudo-t distribution, computed as the mean spectral magnitude divided by the standard error across participants, was used as the initial threshold for family-wise error correction in multiple comparisons.

Anatomical masks

The mask of the bilateral EC was manually delineated on the MNI T1 brain with 1 mm resolution packaged by FSL, using established protocols (Insausti et al., 1998), and the delineation software ITK-SNAP (Version 3.8, www.itksnap.org). The EC mask was then resampled to a 2 mm resolution. The anatomical mask of the bilateral HPC and the MTL were derived from the AAL atlas, an automated anatomical parcellation of the spatially normalized single-subject high-resolution T1 volume provided by the Montreal Neurological Institute (Version: AAL 3v2; <https://www.gin.cnr.fr/en/tools/aal/>) (Rolls et al., 2020).

Spectral analysis of human behavior

FFT analysis was employed to examine participants' behavioral periodicity, following the same procedure used to detect activity periodicity in the EC and HPC. For each participant, a performance vector was constructed from their paths. These vectors were down-sampled into 10° bins from the original 288 conceptual directions (from 0° to 360°), sorted in ascending order, detrended and processed with a Hanning window. This procedure yielded a resampled vector of 36 data points, which was then entered into the FFT analysis. The two-tailed 95th percentile of the spectral magnitude distribution was used as the initial threshold for the family-wise error correction across multiple comparisons.

The “EC-HPC PhaseSync” model

The EC grid-cell spatial code G was simulated using the cosine grating model (O'Keefe and Burgess, 2005; Burgess et al., 2007; Blair et al., 2007; Bush and Schmidt-Hieber, 2018) in Python (version 3.9). At each Greeble location $r = (x, y)$, three cosine gratings, oriented 60° apart, were generated by rotating around the anchor point r according to the rotation vector k , and then linearly summed (Equation 1). The parameter A represents the amplitude of the cosine gratings, $c = (c_x, c_y)$ denotes the spatial phase offset of the grating pattern, and ω indicates the scale ratio (i.e., grid modules). This procedure generated a 45-by-45 grid cell population, with a constant grid orientation maintained across the population.

$$G_r = \sum_{i=1}^3 A \cos(\omega_i(k_{i,x}(r_x - c_x) + k_{i,y}(r_y - c_y))) \quad (1)$$

The path code V of grid cell population was defined as the linear summation of grid codes across locations along the path (Equation 2), where r and q denote the starting location (i.e., Greeble variant) and the ending location (Greeble prototype), and r' indexes the intermediate locations along the path from r to q .

$$V_r = \sum_{r' \in trajectory(r,q)} G_{r'} \quad (2)$$

V exhibited a “planar wave” pattern (Welday et al., 2011; Krupic et al., 2012) when the allocentric direction Φ , defined by $\arctan 2 \left(\frac{q_y - r_y}{q_x - r_x} \right)$, aligned with the primary axes of grid cells (i.e., orientation [1][3][5]; aligned conditions). In contrast, when the allocentric direction Φ_r ,

deviated from the grid axes (i.e., orientation [2][4][6]; ; misaligned conditions), irregular activity patterns emerged from the path code V . Note that the path code V is exactly identical between the allocentric direction Φ and $\Phi + 180^\circ$. In other words, the Greeble space can be tessellated by a vector of orientations over $\psi \in [0, \pi)$ in the orientation domain (Equation 3 [4]).

$$\psi_r = (\arctan2(q_y - r_y, q_x - r_x)) \bmod \pi \quad (3)$$

The neural representations of spatial orientation ψ can be simplified as a two-dimensional vector δ , where each δ_r , representing activity in the 2D orientation space, was derived from the path code V_r by linearly summing the activity of a set of locations R along the path (Equation 4 [4]). The magnitude of δ_r indexed the degree of alignment between the orientation ψ and the grid axes, with larger values indicating closer alignment. In the orientation domain, the δ vector was expected to exhibit a 3-fold periodicity across environmental orientations, characterized by a repeating “aligned-deviated” pattern driven by the periodic structure of the path codes $V(\psi)$ projected from grid cell population, with the spatial phase inherited from grid orientation.

$$\delta_r = \sum_{r \in R_\psi} V_{r,\psi} \quad (4)$$

To construct the goal-directed vector representation C of the HPC for driving movements from self-location towards goal-location, the allocentric directions Φ between the location vector R and the goal location were encoded by the δ periodic structure, with the spatial fold and spatial phase denoted by δ_c and δ_e , respectively (Equation 5 [4]). Then, C was the linearly summation between δ and a Gaussian-based distance term centered at the goal location q (Equation 5 [4]). The distance term is invariant to spatial directions but encodes spatial proximity to the goal. The parameter σ represents the spatial spread, defined by the radius of the Greeble space.

$$C(R) = \cos(\delta_c \times \Phi_R + \delta_e) + \exp\left(-\frac{\|R-q\|^2}{2\sigma^2}\right) \quad (5)$$

Statistical analysis

Voxel-wise clusters from the sinusoidal modulation and spectral analyses were first identified using a cluster-defining threshold of $p < 0.05$ (two-tailed t-test). Cluster significance was determined with cluster-based permutation (5,000 iterations) using FSL randomise (v2.9; Nichols & Holmes, 2002 [4]), with small-volume correction (SVC) applied within the bilateral MTL. Clusters exceeding the 95th percentile of the maximal suprathreshold cluster size were deemed significant. The directional tuning reliability was assessed via a label-shuffling permutation, in which directional labels were randomly permuted 5,000 times to derive a null distribution of spectral power. The 95th percentile of surrogate power defined the uncorrected threshold, and the across-fold 95th percentile served as the FWE-corrected threshold. For the PPC (Vinck et al., 2010 [4]), significance was tested using 5,000 permutations of uniformly distributed random phases ($0-2\pi$) to generate a null distribution for comparison with the observed PPC. For behavioral periodicity, both human and EC-HPC PhaseSync model data were evaluated against shuffled null distributions (5,000 iterations). Multiple comparisons across folds (0–18) were controlled using the maximum-statistic approach. Finally, coherence analyses were performed for EC-HPC and behavior-HPC coupling. Direction-domain signals were band-pass filtered at the 3- and 6-fold using a two-way least-squares FIR filter (eegfilt.m, EEGLAB; Delorme & Makeig, 2004 [4]). EC-HPC coherence was quantified using amplitude-phase coupling (Canolty et al., 2006 [4]), and behavior-HPC coherence was quantified using the phase-lag index (PLI; Stam et al., 2007 [4]). The relationship between EC and HPC phases was evaluated using the circular-circular correlation (Jammalamadaka & Sengupta, 2001 [4]) implemented in the CircStat MATLAB toolbox. For all coherence and phase-coupling analyses, statistical significance was assessed using a non-parametric permutation test. Surrogate datasets were generated by circularly shifting the signal series along the direction axis across all possible offsets (Canolty et al., 2006 [4]), thereby preserving the within-domain phase structure while disrupting consistent phase alignment between signals. Each surrogate dataset

underwent identical filtering and coherence computation to obtain a null distribution. The observed coherence strength was then compared with this distribution using paired t-tests across participants.

Supplemental information

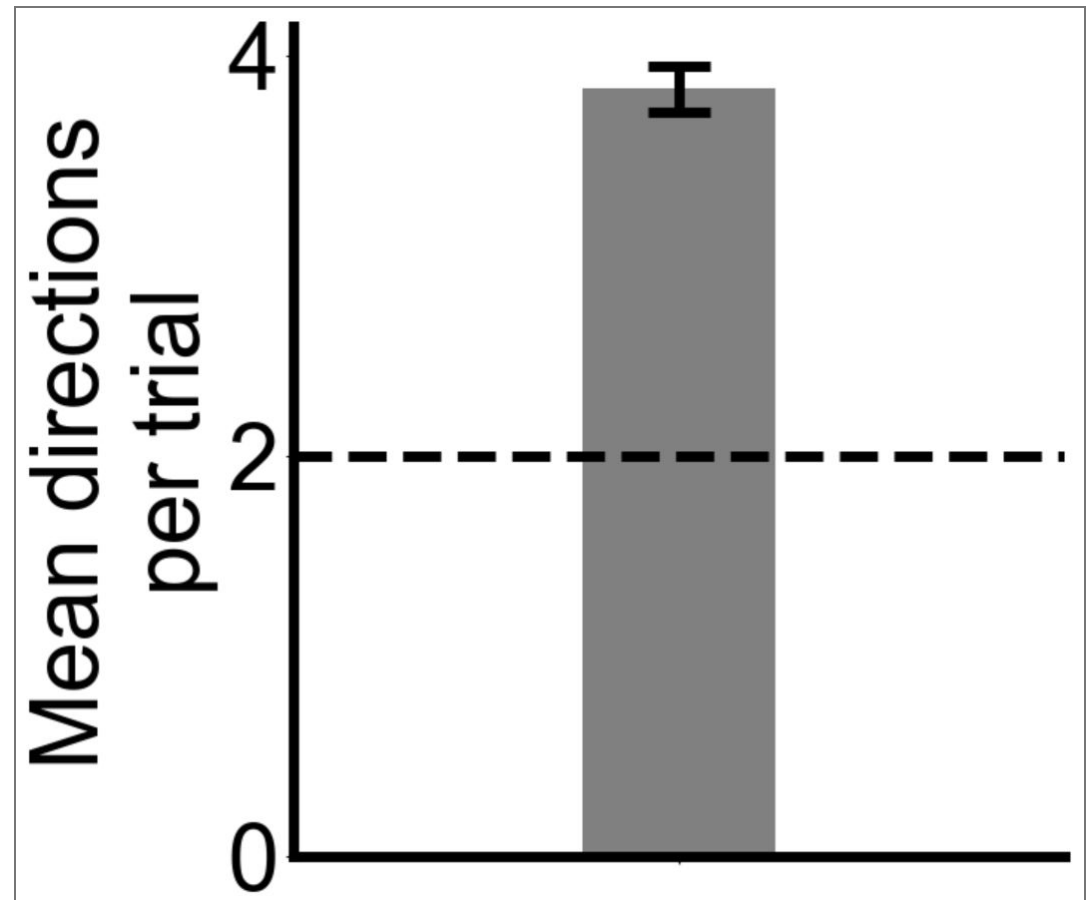


Fig. S1. The averaged number of path directions across all paths (i.e., trial). The original path directions towards East (e.g., key 1 on the response box), West (e.g., key 2), North (e.g., key 3), and South (e.g., key 4) were identified based on three consecutive movement steps within each path. On average, participants moved in 3.8 directions, suggesting a “Radial Adjustment” strategy and excluding the possibility of the “horizontal-vertical movement” strategy, which would result in 2 path directions (black dashed line) per path ($t(32) = 15.76$, $p < 0.001$; two-tailed; Cohen’s $d = 2.78$). The error bars indicate SEM.

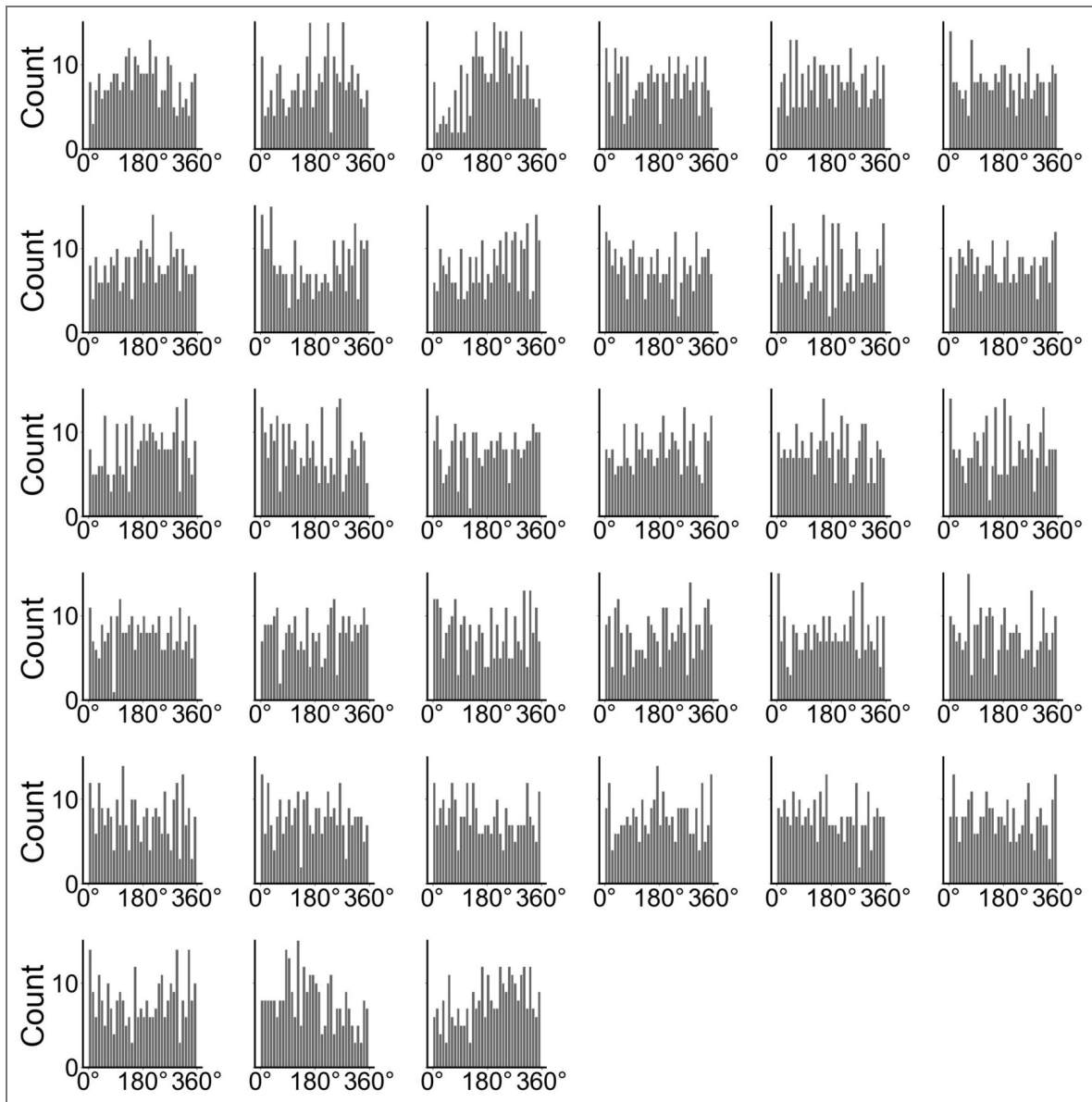


Fig. S2. Histograms of path directions for each participant.

Path directions were binned into 36 bins of 10° each. The uniformity of the distributions was assessed using the Rayleigh test. For all participants, no significant deviations from uniformity were observed (all $p > 0.05$, Bonferroni-corrected across participants). Bar height indicates the number of samples per bin.

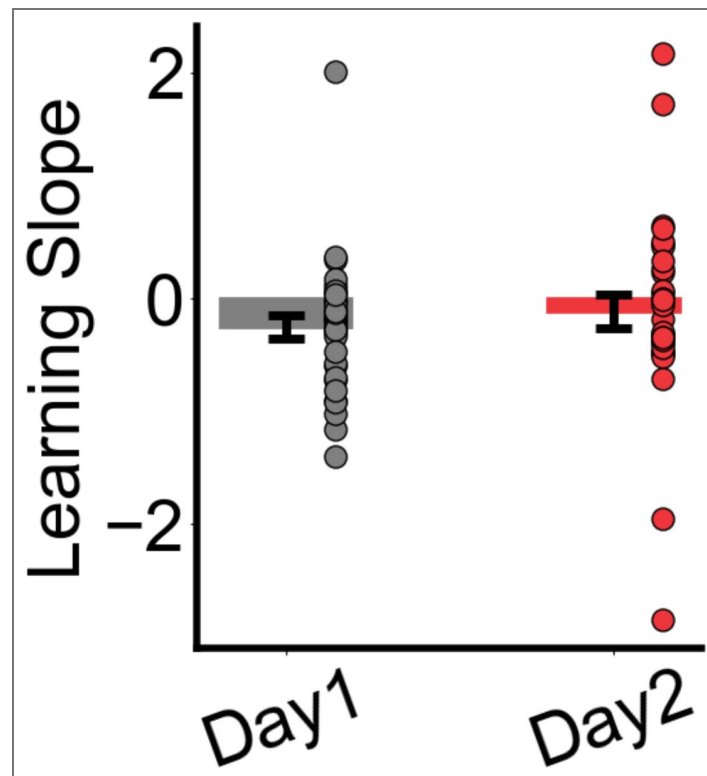


Fig. S3. Learning effect of the object matching task.

The two-day design effectively eliminated the learning effect from the MRI experiment. On day 1, participants' behavioral performance increased as a function of experimental sessions, resulting a significantly negative slope (grey bar, $t(32) = -2.46$, $p = 0.019$, two-tailed; Cohen's $d = 0.44$). In contrast, no significant learning effect was found on day 2 (red bar, $t(32) = -0.74$, $p = 0.462$, two-tailed; Cohen's $d = 0.13$). The dots indicate individual participants. The error bars indicate SEM.

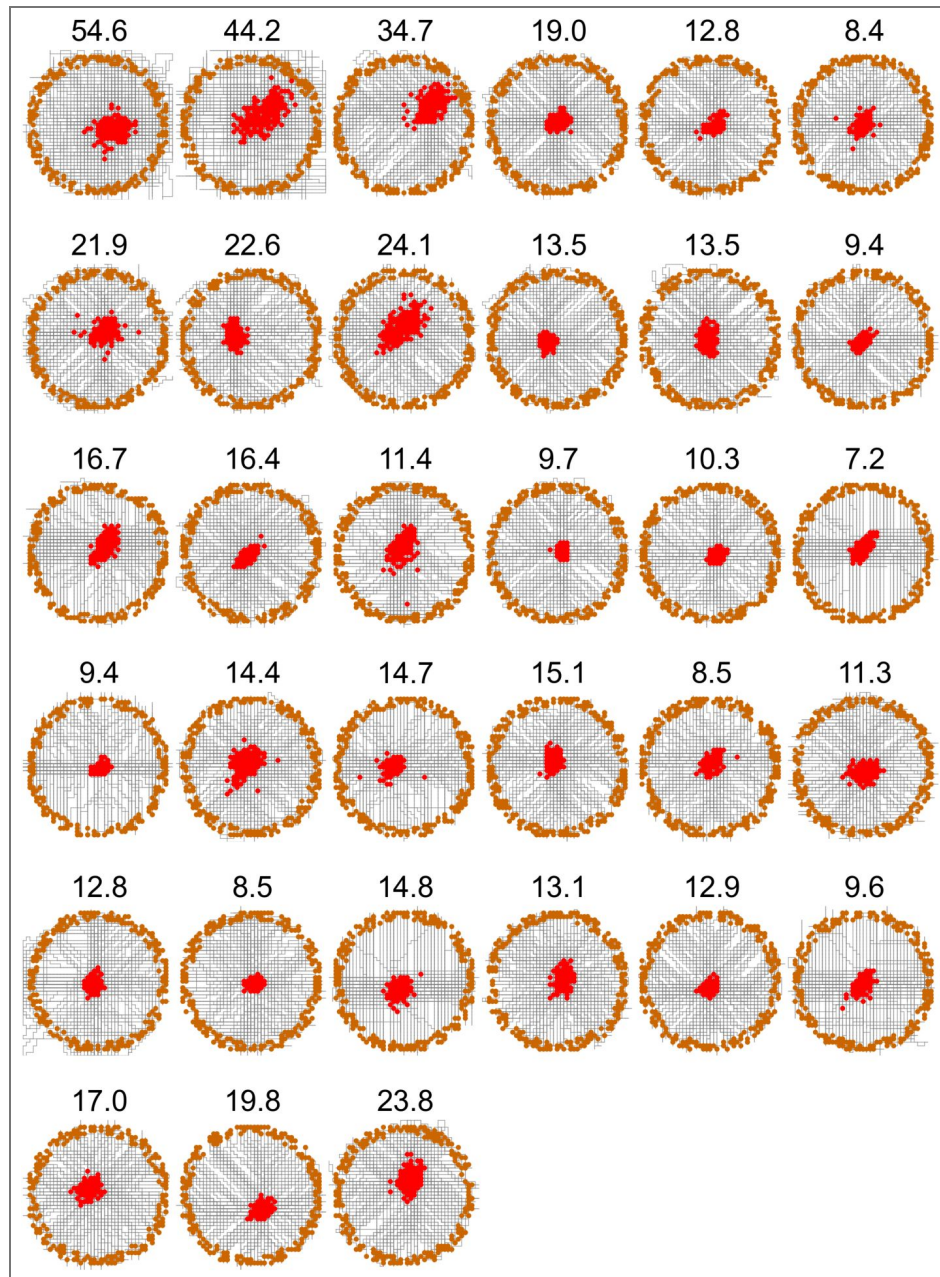


Fig. S4. Movement paths of human participants during Greeble morphing in the MRI experiment.

Orange and red dots represent the starting and ending locations, respectively. The black lines indicate the movement paths. Participants' behavioral performance, calculated as the sum of path length and error size, is shown in the title, with smaller score indicating better performance.

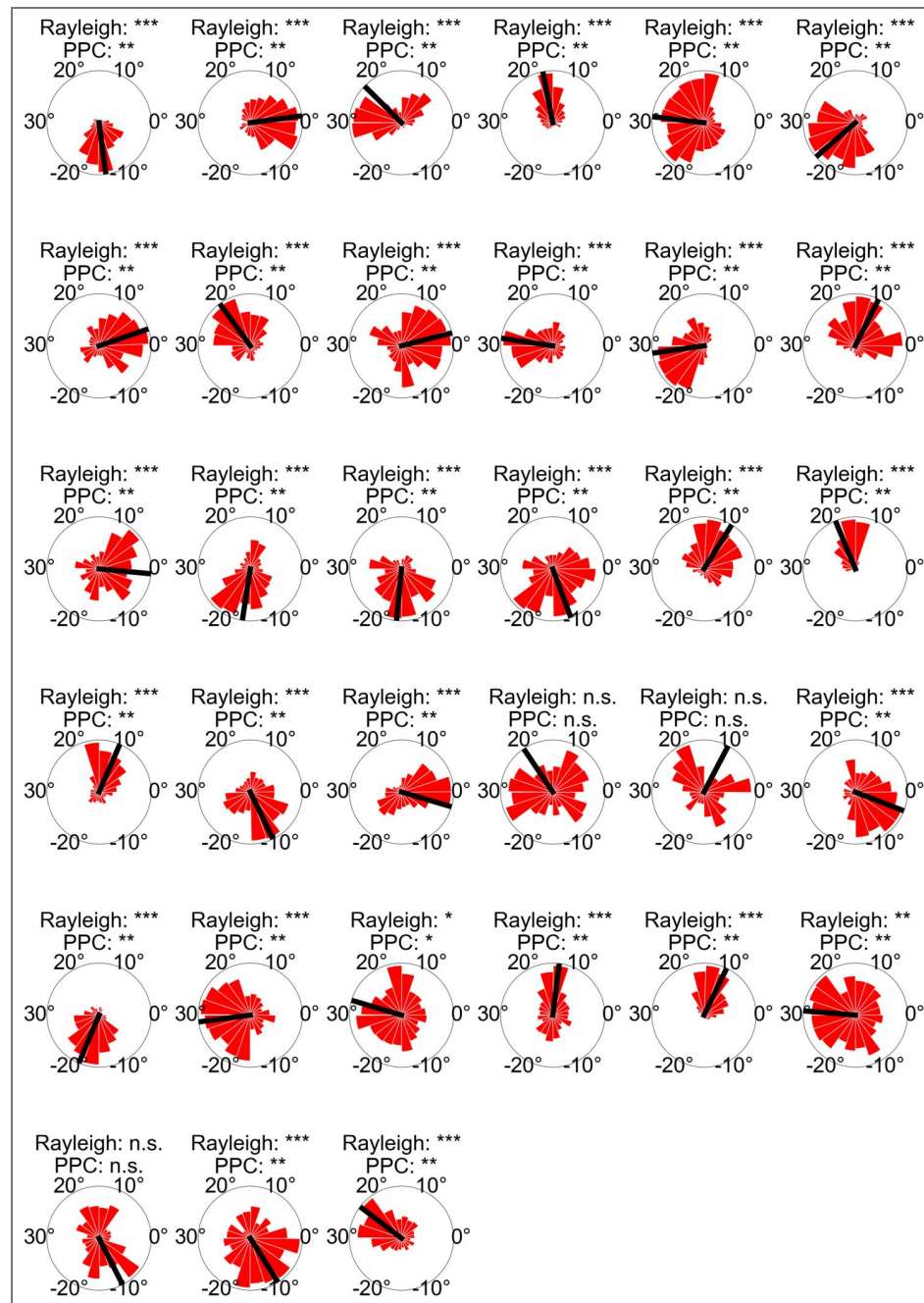


Fig. S5. Distribution of grid orientations in the EC.

Grid orientations ranging from -30° to 30° were estimated for each voxel. The uniformity of grid orientations was assessed using both the Rayleigh test and PPC. Significantly clustering was observed in 30 out of 33 participants, indicated by * (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, Bonferroni-corrected across participants). Bar length indicates the number of voxels. Black line denotes the mean orientation.

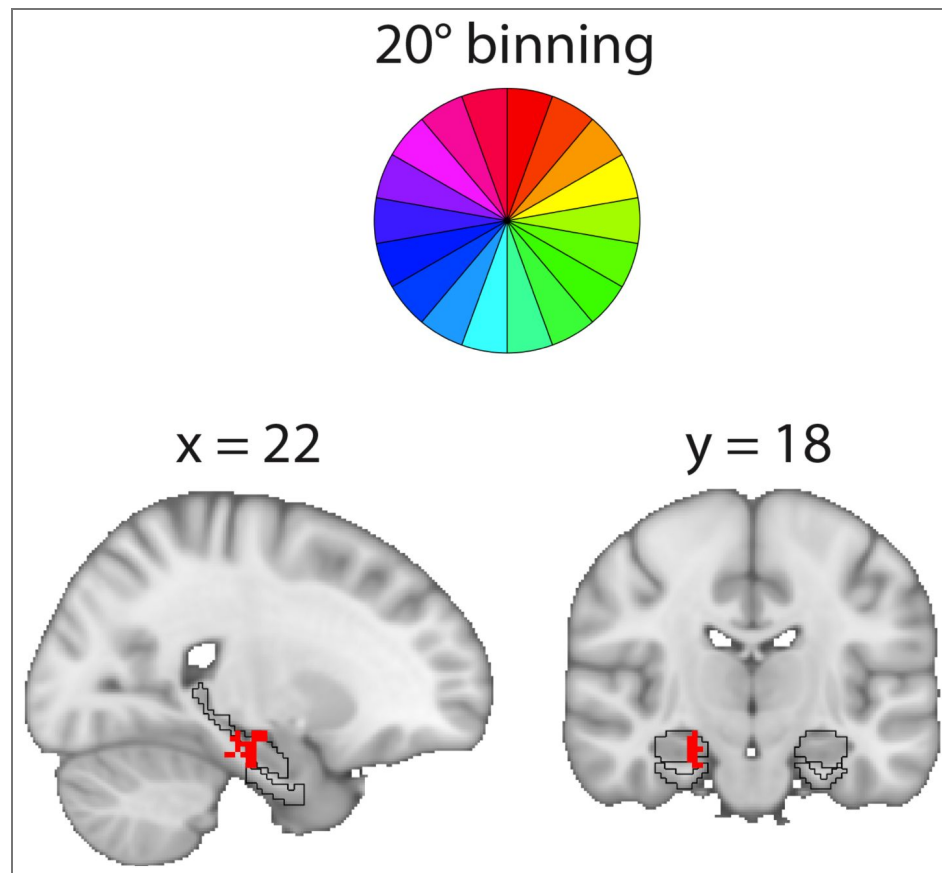


Fig. S6. 3-fold periodicity of the HPC examined using 20° bin.

Significant clusters were revealed in the right HPC (initial threshold: $p = 0.05$, two-tailed; clustered based SVC correction for multiple comparisons: $p < 0.05$; Cohen's $d = 1.18$; Peak MNI coordinate: 22, -24, -14). The black lines indicate the outline of the HPC and EC. Volumetric results are displayed in radiological orientation; numbers above the brain slices indicate MNI coordinates.

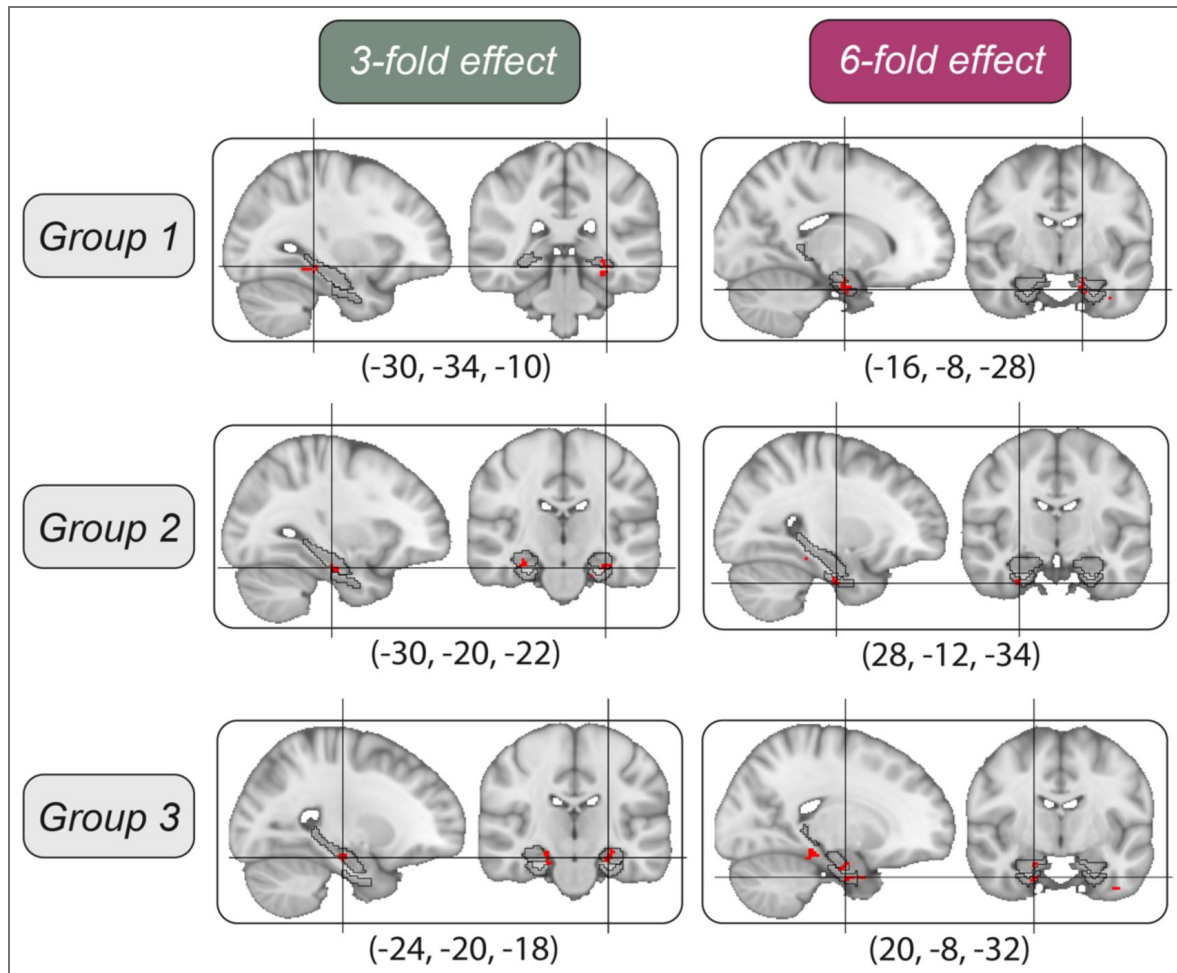


Fig. S7. 3- and 6-fold periodicity of participant groups revealed by spectral analysis.

All individual groups showed significant clusters of 3- and 6-fold periodicity in the HPC and EC, respectively (initial threshold: $p = 0.05$, two-tailed. Cluster-based SVC correction for multiple comparisons: $p < 0.05$). The peak MNI coordinates are shown on the bottom (3- fold: Group 1-3, $t(9) = 8.0$, $t(9) = 7.6$, $t(12) = 10.8$, Cohen's $d = 0.67, 0.87, 1.21$; 6-fold: Group 1-3, $t(9) = 13.8$, $t(9) = 8.7$, $t(12) = 6.7$; Cohen's $d = 0.83, 0.63, 0.96$). The black lines indicate the outline of the HPC and EC. Volumetric results are displayed in radiological orientation; numbers below the brain slices indicate MNI coordinates.

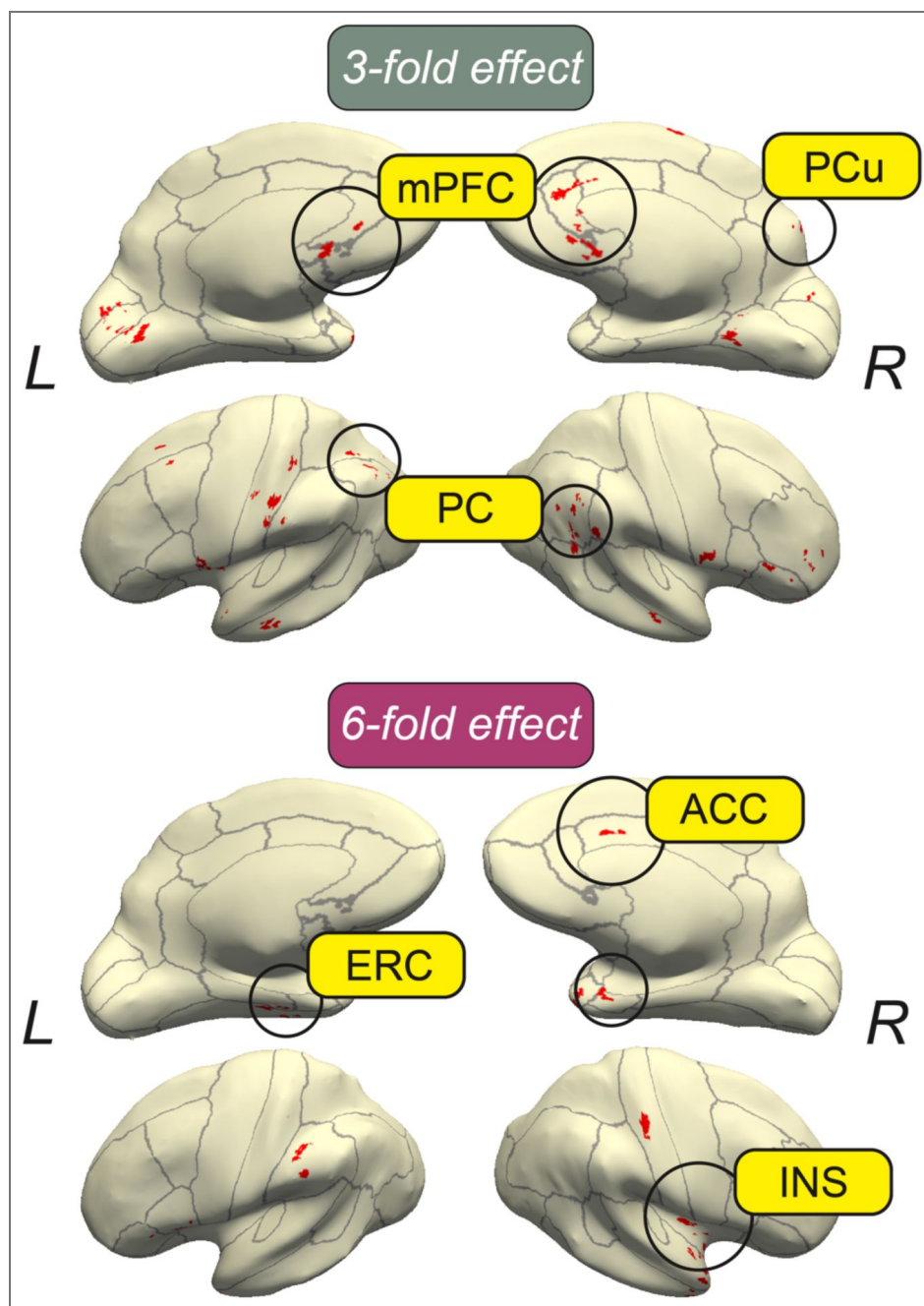


Fig. S8. Whole brain representations of 6- and 3-fold periodicity revealed by spectral analysis.

Significant clusters (Red) within the DMN were identified as representing 3-fold periodicity, whereas 6-fold periodicity revealed clusters within the Saliience Network, including the ACC and INS (initial threshold: $p = 0.05$, two-tailed. Whole brain cluster-based correction for multiple comparisons: $p < 0.05$). The grey lines represent the cortical parcellation based on the DKT Atlas.

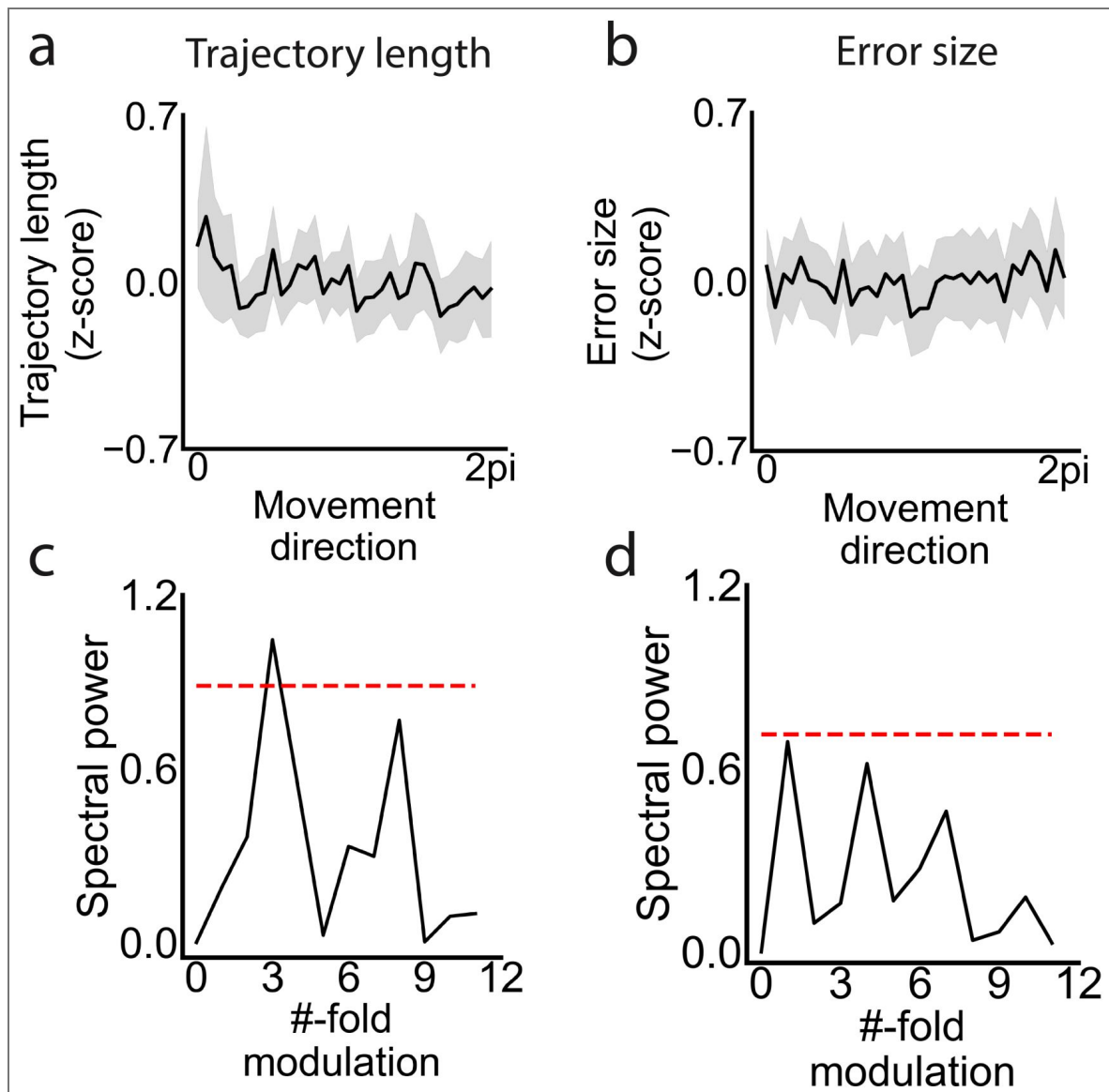


Fig. S9. Spectral analysis of human behavior.

a & b) Behavioral periodicity of path length and error size (deviation from the goal to ending locations). The shaded areas denote SE. c & d) Spectral analysis revealed significant 3-fold power for path length ($p < 0.05$, corrected for multiple comparisons), whereas no significant periodicity was observed for error size ($p > 0.05$). The red dashed line indicates the permutation-based baseline.

Data availability

The analyses in the present study were carried out using custom scripts written in Python (version 3.9), Pygame (version 2.0), Matlab (version 2019b), R (version 4.0), and GNU Bash (version 3.2.57). The neuroimaging and machine learning toolkits employed include FreeSurfer (version 7.1, <https://surfer.nmr.mgh.harvard.edu/>), FSL (version 6.0, <https://fsl.fmrib.ox.ac.uk/fsl/fslwiki>), and Torch (version 1.9, <https://pytorch.org/>). The MRI dataset, behavioral data, and our manually delineated EC mask are available via the Science Data Bank (<https://www.scidb.cn/s/NBriAn>). The original code for the "EC-HPC PhaseSync" model is openly available at (<https://github.com/ZHANGneuro/The-E-H-PhaseSync-Model>). Additional scripts are available upon request from the corresponding author (J.L.).

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Investigation: B.Z., X.G.

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Peer reviews

Reviewer #1 (Public review):

Summary:

Zhang and colleagues examine neural representations underlying abstract navigation in entorhinal cortex (EC) and hippocampus (HC) using fMRI. This paper replicates a previously identified hexagonal modulation of abstract navigation vectors in abstract space in EC in a novel task involving navigating in a conceptual Greeble space. In HC, the authors identify a three-fold signal of the navigation angle. They also use a novel analysis technique (spectral analysis) to look at spatial patterns in these two areas and identify phase coupling between HC and EC. Interestingly, the three-fold pattern identified in the hippocampus explains quirks in participants' behavior where navigation performance follows a three-fold periodicity. Finally, the authors propose a EC-HPC PhaseSync Model to understand how the EC and HC construct cognitive maps. The wide array and creativity of the techniques used is impressive but because of their unique nature, the paper would benefit from more details on how some of these techniques were implemented.

Comments on revisions:

Most of my concerns were adequately addressed, and I believe the paper is greatly improved. I have two more points. I noticed that the legend for Figure 4 still refers to some components of the previous figure version, this should be updated to reflect the current version of the figure. I also think the paper would benefit from more details regarding some of the analyses. Specifically, the phase-amplitude coupling analysis should have a section in the methods which should be sure to clarify how the BOLD signals were reconstructed.

<https://doi.org/10.7554/eLife.107517.2.sa2>

Reviewer #2 (Public review):

The authors report results from behavioral data, fMRI recordings, and computer simulations during a conceptual navigation task. They report 3-fold symmetry in behavioral and simulated model performance, 3-fold symmetry in hippocampal activity, and 6-fold symmetry in entorhinal activity (all as a function of movement directions in conceptual space). The analyses seem thoroughly done, and the results and simulations are very interesting.

<https://doi.org/10.7554/eLife.107517.2.sa1>

Author response:

The following is the authors' response to the original reviews.

| **Public Reviews:**

Reviewer #1 (Public review):

Summary:

Zhang and colleagues examine neural representations underlying abstract navigation in the entorhinal cortex (EC) and hippocampus (HC) using fMRI. This paper replicates a previously identified hexagonal modulation of abstract navigation vectors in abstract space in EC in a novel task involving navigating in a conceptual Greeble space. In HC, the authors claim to identify a three-fold signal of the navigation angle. They also use a novel analysis technique (spectral analysis) to look at spatial patterns in these two areas and identify phase coupling between HC and EC. Finally, the authors propose an EC-HPC PhaseSync Model to understand how the EC and HC construct cognitive maps. While the wide array of techniques used is impressive and their creativity in analysis is admirable, overall, I found the paper a bit confusing and unconvincing. I recommend a significant rewrite of their paper to motivate their methods and clarify what they actually did and why. The claim of three-fold modulation in HC, while potentially highly interesting to the community, needs more background to motivate why they did the analysis in the first place, more interpretation as to why this would emerge in biology, and more care taken to consider alternative hypotheses seeped in existing models of HC function. I think this paper does have potential to be interesting and impactful, but I would like to see these issues improved first.

General comments:

(1) Some of the terminology used does not match the terminology used in previous relevant literature (e.g., sinusoidal analysis, 1D directional domain).

We thank the reviewer for this valuable suggestion, which helps to improve the consistency of our terminology with previous literature and to reduce potential ambiguity. Accordingly, we have replaced “sinusoidal analysis” with “sinusoidal modulation” (Doeller et al., 2010; Bao et al., 2019; Raithel et al., 2023) and “1D directional domain” with “angular domain of path directions” throughout the manuscript.

(2) Throughout the paper, novel methods and ideas are introduced without adequate explanation (e.g., the spectral analysis and three-fold periodicity of HC).

We thank the reviewer for raising this important point. In the revised manuscript, we have substantially extended the Introduction (paragraphs 2–4) to clarify our hypothesis, explicitly explaining why the three primary axes of the hexagonal grid cell code may manifest as vector fields. We have also revised the first paragraph of the “3-fold periodicity in the HPC” section in the Results to clarify the rationale for using spectral analysis. Please refer to our responses to comment 2 and 3 below for details.

Reviewer #2 (Public review):

The authors report results from behavioral data, fMRI recordings, and computer simulations during a conceptual navigation task. They report 3-fold symmetry in behavioral and simulated model performance, 3-fold symmetry in hippocampal activity, and 6-fold symmetry in entorhinal activity (all as a function of movement directions in conceptual space). The analyses are thoroughly done, and the results and simulations are very interesting.

We sincerely thank the reviewer for the positive and encouraging comments on our study.

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

(1) This paper has quite a few spelling and grammatical mistakes, making it difficult to understand at times.

We apologize for the wordings and grammatical errors. We have thoroughly re-read and carefully edited the entire manuscript to correct typographical and grammatical errors, ensuring improved clarity and readability.

(2) Introduction - It's not clear why the three primary axes of hexagonal grid cell code would manifest as vector fields.

We thank the reviewer for raising this important point. In the revised Introduction (paragraphs 2, 3, and 4), we now explicitly explain the rationale behind our hypothesis that the three primary axes of the hexagonal grid cell code manifest as vector fields.

In paragraph 2, we present empirical evidence from rodent, bat, and human studies demonstrating that mental simulation of prospective paths relies on vectorial representations in the hippocampus (Sarel et al., 2017; Ormond and O'Keefe, 2022; Muhle-Karbe et al., 2023).

In paragraphs 3 and 4, we introduce our central hypothesis: vectorial representations may originate from population-level projections of entorhinal grid cell activity, based on three key considerations:

- (1) The EC serves as the major source of hippocampal input (Witter and Amaral, 1991; van Groen et al., 2003; Garcia and Buffalo, 2020).
- (2) Grid codes exhibit nearly invariant spatial orientations (Hafting et al., 2005; Gardner et al., 2022), which makes it plausible that their spatially periodic activity can be detected using fMRI.
- (3) A model-based inference: for example, in the simplest case, when one mentally simulates a straight pathway aligned with the grid orientation, a subpopulation of grid cells would be activated. The resulting population activity would form a near-perfect vectorial representation, with constant activation strength along the path. In contrast, if the simulated path is misaligned with the grid orientation, the population response becomes a distorted vectorial code. Consequently, simulating all possible straight paths spanning 0°–360° results in 3-fold periodicity in the activity patterns—due to the 180° rotational symmetry of the hexagonal grid, orientations separated by 180° are indistinguishable.

We therefore speculate that vectorial representations embedded in grid cell activity exhibit 3-fold periodicity across spatial orientations and serve as a periodic structure to represent spatial direction. Supporting this view, reorientation paradigms in both rodents and young children have shown that subjects search equally in two opposite directions, reflecting successful orientation encoding but a failure to integrate absolute spatial direction (Hermer and Spelke, 1994; Julian et al., 2015; Gallistel, 2017; Julian et al., 2018).

(3) It took me a few reads to understand what the spectral analysis was. After understanding, I do think this is quite clever. However, this paper needs more motivation to understand why you are performing this analysis. E.g., why not just take the average regressor at the 10°, 70°, etc. bins and compare it to the average regressor at 40°, 100° bins? What does the Fourier transform buy you?

We are sorry for the confusion. we outline the rationale for employing Fast Fourier Transform (FFT) analysis to identify neural periodicity. In the revised manuscript, we have added these clarifications into the first paragraph of the “3-fold periodicity in the HPC” subsection in the Results.

First, FFT serves as an independent approach to cross-validate the sinusoidal modulation results, providing complementary evidence for the 6-fold periodicity in EC and the 3-fold periodicity in HPC.

Second, FFT enables unbiased detection of multiple candidate periodicities (e.g., 3–7-fold) simultaneously without requiring prior assumptions about spatial phase (orientation). By contrast, directly comparing “aligned” versus “misaligned” angular bins (e.g., $10^\circ/70^\circ$ vs. $40^\circ/100^\circ$) would implicitly assume knowledge of the phase offset, which was not known a priori.

Finally, FFT uniquely allows periodicity analysis of behavioral performance, which is not feasible with standard sinusoidal GLM approaches. This methodological consistency makes it possible to directly compare periodicities across neural and behavioral domains.

(4) A more minor point: at one point, you say it's a spectral analysis of the BOLD signals, but the methods description makes it sound like you estimated regressors at each of the bins before performing FFT. Please clarify.

We apologize for the confusion. In our manuscript, we use the term spectral analysis to distinguish this approach from sinusoidal modulation analysis. Conceptually, our spectral analysis involves a three-level procedure:

(1) First level: We estimated direction-dependent activity maps using a general linear model (GLM), which included 36 regressors corresponding to path directions, down-sampled in 10° increments.

(2) Second level: We applied a Fast Fourier Transform (FFT) to the direction-dependent activity maps derived from the GLM to examine the spectral magnitude of potential spatial periodicities.

(3) Third level: We conducted group-level statistical analyses across participants to assess the consistency of the observed periodicities.

We have revised the “Spectral analysis of MRI BOLD signals” subsection in the Methods to clarify this multi-level procedure.

(5) Figure 4a:

Why do the phases go all the way to 2π if periodicity is either three-fold or six-fold?

When performing correlation between phases, you should perform a circular-circular correlation instead of a Pearson's correlation.

We thank the reviewer for raising this important point. In the original Figure 4a, both EC and HPC phases spanned $0-2\pi$ because their sinusoidal phase estimates were projected into a common angular space by scaling them according to their symmetry factors (i.e., multiplying the 3-fold phase by 3 and the 6-fold phase by 6), followed by taking the modulo 2π . However, this projection forced signals with distinct intrinsic periodicities (120° vs. 60° cycles) into a shared 360° space, thereby distorting their relative angular distances and disrupting the one-to-one correspondence between physical directions and phase values. Consequently, this transformation could bias the estimation of their phase relationship.

In the revised analysis and Figure 4a, we retained the original phase estimates derived from the sinusoidal modulation within their native periodic ranges ($0-120^\circ$ for 3-fold and $0-60^\circ$ for 6-fold) by applying modulo operations directly. Following your suggestion, the relationship between EC and HPC phases was then quantified using circular–circular correlation (Jammalamadaka & Sengupta, 2001), as implemented in the CircStat MATLAB toolbox. This

updated analysis avoids the rescaling artifact and provides a statistically stronger and conceptually clearer characterization of the phase correspondence between EC and HPC.

(6) Figure 4d needs additional clarification:

Phase-locking is typically used to describe data with a high temporal precision. I understand you adopted an EEG analysis technique to this reconstructed fMRI time-series data, but it should be described differently to avoid confusion. This needs additional control analyses (especially given that 3 is a multiple of 6) to confirm that this result is specific to the periodicities found in the paper.

We thank the reviewer for this insightful comment. We have extensively revised the description of the Figure 4 to avoid confusion with EEG-based phase-locking techniques. The revised text now explicitly clarifies that our approach quantifies spatial-domain periodic coupling across path directions, rather than temporal synchronization of neural signals.

To further address the reviewer's concern about potential effects of the integer multiple relationship between the 3-fold HPC and 6-fold EC periodicities, we additionally performed two control analyses using the 9-fold and 12-fold EC components, both of which are also integer multiples of the 3-fold HPC periodicity. Neither control analysis showed significant coupling ($p > 0.05$), confirming that the observed 3-fold–6-fold coupling was specific and not driven by their harmonic relationship.

The description of the revised Figure 4 has been updated in the “Phase Synchronization Between HPC and EC Activity” subsection of the Results.

(7) Figure 5a is misleading. In the text, you say you test for propagation to egocentric cortical areas, but I don't see any analyses done that test this. This feels more like a possible extension/future direction of your work that may be better placed in the discussion.

We are sorry for the confusion. Figure 5a was intended as a hypothesis-driven illustration to motivate our analysis of behavioral periodicity based on participants' task performance. However, we agree with the reviewer that, on its own, Figure 5a could be misleading, as it does not directly present supporting analyses.

To provide empirical support for the interpretation depicted in Figure 5a, we conducted a whole-brain analysis (Figure S8), which revealed significant 3-fold periodic signals in egocentric cortical regions, including the parietal cortex (PC), precuneus (PCU), and motor regions.

To avoid potential misinterpretation, we have revised the main text to include these results and explicitly referenced Figure S8 in connection with Figure 5a.

The updated description in the “3-fold periodicity in human behavior” subsection in the Results is as follows:

“Considering the reciprocal connectivity between the medial temporal lobe (MTL), where the EC and HPC reside, and the parietal cortex implicated in visuospatial perception and action, together with the observed 3-fold periodicity within the DMN (including the PC and PCu; Fig. S8), we hypothesized that the 3-fold periodic representations of path directions extend beyond the MTL to the egocentric cortical areas, such as the PC, thereby influencing participants' visuospatial task performance (Fig. 5a)”.

Additionally, Figure 5a has been modified to more clearly highlight the hypothesized link between activity periodicity and behavioral periodicity, rather than suggesting a direct anatomical pathway.

(8) PhaseSync model: I am not an expert in this type of modeling, so please put a lower weight on this comment (especially compared to some of the other reviewers). While the PhaseSync model seems interesting, it's not clear from the discussion how this compares to current models. E.g., Does it support them by adding the three-fold HC periodicity? Does it demonstrate that some of them can't be correct because they don't include this three-fold periodicity?

We thank the reviewer for the insightful comment regarding the PhaseSync model. We agree that further clarifying its relationship to existing computational frameworks is important.

The EC–HPC PhaseSync model is not intended to replace or contradict existing grid–place cell models of navigation (e.g., Bicanski and Burgess, 2019; Whittington et al., 2020; Edvardsen et al., 2020). Instead, it offers a hierarchical extension by proposing that vectorial representations in the hippocampus emerge from the projections of periodic grid codes in the entorhinal cortex. Specifically, the model suggests that grid cell populations encode integrated path information, forming a vectorial gradient toward goal locations.

To simplify the theoretical account, our model was implemented in an idealized square layout. In more complex real-world environments, hippocampal 3-fold periodicity may interact with additional spatial variables, such as distance, movement speed, and environmental boundaries.

We have revised the final two paragraphs of the Discussion to clarify this conceptual framework and emphasize the importance of future studies in exploring how periodic activity in the EC–HPC circuit interacts with environmental features to support navigation.

Reviewer #2 (Recommendations for the authors):

(1) Please show a histogram of movement direction sampling for each participant.

We thank the reviewer for this helpful suggestion. We have added a new supplementary figure (Figure S2) showing histograms of path direction sampling for each participant (36 bins of 10°). The figure is also included. Rayleigh tests for circular uniformity revealed no significant deviations from uniformity (all p s > 0.05, Bonferroni-corrected across participants), confirming that path directions were sampled evenly across 0°–360°.

(2) Why didn't you use participants' original trajectories (instead of the trajectories inferred from the movement start and end points) for the hexadirectional analyses?

In our paradigm, participants used two MRI-compatible 2-button response boxes (one for each hand) to adjust the two features of the greebles. As a result, the raw adjustment path contained only four cardinal directions (up, down, left, right). If we were to use the raw stepwise trajectories, the analysis would be restricted to these four directions, which would severely limit the angular resolution. By instead defining direction as the vector from the start to the end position in feature space, we can expand the effective range of directions to the full 0–360°. This approach follows previous literature on abstract grid-like coding in humans (e.g., Constantinescu et al., 2016), where direction was similarly defined by the relative change between two feature dimensions rather than the literal stepwise path. We have added this clarification in the “Sinusoidal modulation” subsection of the revised method.

(3) Legend of Figure 2: the statement "localizing grid cell activity" seems too strong because it is still not clear whether hexadirectional signals indeed result from grid-cell activity (e.g., Bin Khalid et al., eLife, 2024). I would suggest rephrasing this statement (here and elsewhere).

Thank you for this helpful suggestion. We have removed the statement “localizing grid cell activity” to avoid ambiguity and revised the legend of Figure 2a to more explicitly highlight its main purpose—defining how path directions and the aligned/misaligned conditions were constructed in the 6-fold modulation. We have also modified similar expressions throughout the manuscript to ensure consistency and clarity.

(4) Legend of Figure 2: “cluster-based SVC correction for multiple comparisons” - what is the small volume you are using for the correction? Bilateral EC?

For both Figure 2 and Figure 3, the anatomical mask of the bilateral medial temporal lobe (MTL), as defined by the AAL atlas, was used as the small volume for correction. This has been clarified in the revised Statistical Analysis section of the Methods as “... with small-volume correction (SVC) applied within the bilateral MTL”.

(5) Legend of Figure 2: “ROI-based analysis” - what kind of ROI are you using? “corrected for multiple comparisons” - which comparisons are you referring to? Different symmetries and also the right/left hemisphere?

In Figure 2b, the ROI was defined as a functional mask derived from the significant activation cluster in the right entorhinal cortex (EC). Since no robust clusters were observed in the left EC, the functional ROI was restricted to the right hemisphere. We indeed included Figure 2c to illustrate this point; however, we recognize that our description in the text was not sufficiently clear.

Regarding the correction for multiple comparisons, this refers specifically to the comparisons across different rotational symmetries (3-, 4-, 5-, 6-, and 7-fold). Only the 6-fold symmetry survived correction, whereas no significant effects were detected for the other symmetries.

We have clarified these points in the “6-fold periodicity in the EC” subsection of the result as “... The ROI was defined as a functional mask of the right EC identified in the voxel-based analysis and further restricted within the anatomical EC. These analyses revealed significant periodic modulation only at 6-fold (Figure 2c; $t(32) = 3.56$, $p = 0.006$, two-tailed, corrected for multiple comparisons across rotational symmetries; Cohen’s $d = 0.62$) ...”.

We have also revised the “3-fold periodicity in the HPC” subsection of the result as “... ROI analysis, using a functional mask of the HPC identified in the spectral analysis and further restricted within the anatomical HPC, indicated that HPC activity selectively fluctuated at 3-fold periodicity (Figure 3e; $t(32) = 3.94$, $p = 0.002$, corrected for multiple comparisons across rotational symmetries; Cohen’s $d = 0.70$) ...”.

(6) Figure 2d: Did you rotationally align 0{degree sign} across participants? Please state explicitly whether (or not) 0{degree sign} aligns with the x-axis in Greeble space.

We thank the reviewer for this helpful question. Yes, before reconstructing the directional tuning curve in Figure 2d, path directions were rotationally aligned for each participant by subtracting the participant-specific grid orientation (Φ) estimated from the independent dataset (odd sessions). We have now made this description explicit in the revised manuscript in the “6-fold periodicity in the EC” subsection of the Results, stating “... To account for individual difference in spatial phase, path directions were calibrated by subtracting the participant-specific grid orientation estimated from the odd sessions ...”.

(7) Clustering of grid orientations in 30 participants: What does “Bonferroni corrected” refer to? Also, the Rayleigh test is sensitive to the number of voxels - do you obtain the same results when using pair-wise phase consistency?

“Bonferroni corrected” here refers to correction across participants. We have clarified this in the first paragraph of the “6-fold periodicity in the EC” subsection of the Result and in the

legend of Supplementary Figure S5 as “Bonferroni-corrected across participants.”

To examine whether our findings were sensitive to the number of voxels, we followed the reviewer’s guidance to compute pairwise phase consistency (PPC; Vinck et al., 2010) for each participant. The PPC results replicated those obtained with the Rayleigh test. We have updated the new results into the Supplementary Figure S5. We also updated the “Statistical Analysis” subsection of the Methods to describe PPC as “For the PPC (Vinck et al., 2010), significance was tested using 5,000 permutations of uniformly distributed random phases ($0-2\pi$) to generate a null distribution for comparison with the observed PPC”.

(8) 6-fold periodicity in the EC: Do you compute an average grid orientation across all EC voxels, or do you compute voxel-specific grid orientations?

Following the protocol originally described by Doeller et al. (2010), we estimated voxel-wise grid orientations within the EC and then obtained a participant-specific orientation by averaging across voxels within a hand-drawn bilateral EC mask. The procedure is described in detail in the “Sinusoidal modulation” subsection of the Methods.

(9) Hand-drawn bilateral EC mask: What was your procedure for drawing this mask? What results do you get with a standard mask, for example, from Freesurfer or SPM? Why do you perform this analysis bilaterally, given that the earlier analysis identified 6-fold symmetry only in the right EC? What do you mean by “permutation corrected for multiple comparisons”?

We thank the reviewer for raising these important methodological points. To our knowledge, no standard volumetric atlas provides an anatomically defined entorhinal cortex (EC) mask. For example, the built-in Harvard–Oxford cortical structural atlas in FSL contains only a parahippocampal region that encompasses, but does not isolate, the EC. The AAL atlas likewise does not contain an EC region. In FreeSurfer, an EC label is available, but only in the fsaverage surface space, which is not directly compatible with MNI-based volumetric group-level analyses.

Therefore, we constructed a bilateral EC mask by manually delineating the EC according to the detailed anatomical landmarks described by Insausti et al. (1998). Masks were created using ITK-SNAP (Version 3.8, www.itksnap.org). For transparency and reproducibility, the mask has been made publicly available at the Science Data Bank (link: <https://www.scidb.cn/s/NBriAn>), as indicated in the revised Data and Code availability section.

Regarding the use of a bilateral EC mask despite voxel-wise effects being strongest in the right EC. First, we did not have any a priori hypothesis regarding laterality of EC involvement before performing analyses. Second, previous studies estimated grid orientation using a bilateral EC mask in their sinusoidal analyses (Doeller et al., 2010; Constantinescu et al., 2016; Bao et al., 2019; Wagner et al., 2023; Raithel et al., 2023). We therefore followed this established approach to estimate grid orientation.

By “permutation corrected for multiple comparisons” we refer to the family-wise error correction applied to the reconstructed directional tuning curves (Figure 2d for the EC, Figure 3f for the HPC). Specifically, directional labels were randomly shuffled 5,000 times, and an FFT was applied to each shuffled dataset to compute spectral power at each fold. This procedure generated null distributions of spectral power for each symmetry. For each fold, the 95th percentile of the maximal power across permutations was used as the uncorrected threshold. To correct across folds, the 95th percentile of the maximal suprathreshold power across all symmetries was taken as the family-wise error–corrected threshold. We have clarified this procedure in the revised “Statistical Analysis” subsection of the Methods.

(10) Figures 3b and 3d: Why do different hippocampal voxels show significance for the sinusoidal versus spectral analysis? Shouldn't the analyses be redundant and, thus, identify the same significant voxels?

We thank the reviewer for this insightful question. Although both sinusoidal modulation and spectral analysis aim to detect periodic neural activity, the two approaches are methodologically distinct and are therefore not expected to identify exactly the same significant voxels.

Sinusoidal modulation relies on a GLM with sine and cosine regressors to test for phase-aligned periodicity (e.g., 3-fold or 6-fold), calibrated according to the estimated grid orientation. This approach is highly specific but critically depends on accurate orientation estimation. In contrast, spectral analysis applies Fourier decomposition to the directional tuning profile, enabling the detection of periodic components without requiring orientation calibration.

Accordingly, the two analyses are not redundant but complementary. The FFT approach allows for an unbiased exploration of multiple candidate periodicities (e.g., 3–7-fold) without predefined assumptions, thereby providing a critical cross-validation of the sinusoidal GLM results. This strengthens the evidence for 6-fold periodicity in EC and 3-fold periodicity in HPC. Furthermore, FFT uniquely facilitates the analysis of periodicities in behavioral performance data, which is not feasible with standard sinusoidal GLM approaches. This methodological consistency enables direct comparison of periodicities across neural and behavioral domains.

Additionally, the anatomical distributions of the HPC clusters appear more similar between Figure 3b and Figure 3d after re-plotting Figure 3d using the peak voxel coordinates ($x = -24$, $y = -18$), which are closer to those used for Figure 3b ($x = -24$, $y = -20$), as shown in the revised Figure 3.

Taken together, the two analyses serve distinct but complementary purposes.

(11) 3-fold sinusoidal analysis in hippocampus: What kind of small volume are you using to correct for multiple comparisons?

We thank the reviewer for this comment. The same small volume correction procedure was applied as described in R4. Specifically, the anatomical mask of the bilateral medial temporal lobe (MTL), as defined by the AAL atlas, was used as the small volume for correction. This procedure has been clarified in the revised Statistical Analysis section of the Methods as following: “... with small-volume correction (SVC) applied within the bilateral MTL.”

(12) Figure S5: “right HPC” – isn't the cluster in the left hippocampus?

We are sorry for the confusion. The brain image was present in radiological orientation (i.e., the left and right orientations are flipped). We also checked the figure and confirmed that the cluster shown in the original Figure S5 (i.e., Figure S6 in the revised manuscript) is correctly labeled as the right hippocampus, as indicated by the MNI coordinate ($x = 22$), where positive x values denote the right hemisphere. To avoid potential confusion, we have explicitly added the statement “Volumetric results are displayed in radiological orientation” to the figure legends of all volume-based results.

(13) Figure S5: Why are the significant voxels different from the 3-fold symmetry analysis using 10{degree sign} bins?

As shown in R10, the apparent differences largely reflect variation in MNI coordinates. After adjusting for display coordinates, the anatomical locations of the significant clusters are in

fact highly similar between the 10°-binned (Figure 3d, shown above) and the 20°-binned results (Figure S6).

Although both analyses rely on sinusoidal modulation, they differ in the resolution of the input angular bins (10° vs. 20°). Combined with the inherent noise in fMRI data, this makes it unlikely that the two approaches would yield exactly the same set of significant voxels. Importantly, both analyses consistently reveal robust 3-fold periodicity in the hippocampus, indicating that the observed effect is not dependent on angular bin size.

(14) Figure 4a and corresponding text: What is the unit? Phase at which frequency? Are you using a circular-circular correlation to test for the relationship?

We thank the reviewer for raising this important point. In the revised manuscript, we have clarified that the unit of the phase values is radians, corresponding to the 6-fold periodic component in the EC and the 3-fold periodic component in the HPC. In the original Figure 4a, both EC and HPC phases—estimated from sinusoidal modulation—were analyzed using Pearson correlation. We have since realized issues with this approach, as also noted R5 to Reviewer #1.

In the revised analysis and Figure 4a (as shown above), we re-evaluated the relationship between EC and HPC phases using a circular–circular correlation (Jammalamadaka & Sengupta, 2001), implemented in the CircStat MATLAB toolbox. The “Phase synchronization between the HPC and EC activity” subsection of the Result has been accordingly updated as following:

“To examine whether the spatial phase structure in one region could predict that in another, we tested whether the orientations of the 6-fold EC and 3-fold HPC periodic activities, estimated from odd-numbered sessions using sinusoidal modulation with rotationally symmetric parameters (in radians), were correlated across participants. A cross-participant circular–circular correlation was conducted between the spatial phases of the two areas to quantify the spatial correspondence of their activity patterns (EC: purple dots; HPC: green dots) (Jammalamadaka & Sengupta, 2001). The analysis revealed a significant circular correlation (Figure 4a; $r = 0.42$, $p < 0.001$) ...”.

In the “Statistical analysis” subsection of the method:

“... The relationship between EC and HPC phases was evaluated using the circular–circular correlation (Jammalamadaka & Sengupta, 2001) implemented in the CircStat MATLAB toolbox ...”.

(15) Paragraph following “We further examined amplitude-phase coupling...” - please clarify what data goes into this analysis.

We thank the reviewer for this helpful comment. In this analysis, the input data consisted of hippocampal (HPC) phase and entorhinal (EC) amplitude, both extracted using the Hilbert transform from the reconstructed BOLD signals of the EC and HPC derived through sinusoidal modulation. We have substantially revised the description of the amplitude–phase coupling analysis in the third paragraph of the “Phase Synchronization Between HPC and EC Activity” subsection of the Results to clarify this procedure.

(16) Alignment between EC 6-fold phases and HC 3-fold phases: Why don't you simply test whether the preferred 6-fold orientations in EC are similar to the preferred 3-fold phases in HC? The phase-amplitude coupling analyses seem sophisticated but are complex, so it is somewhat difficult to judge to what extent they are correct.

We thank the reviewer for this thoughtful comment. We employed two complementary analyses to examine the relationship between EC and HPC activity. In the revised Figure 4 (as

shown in Figure 4 for Reviewer #1), Figure 4a provides a direct and intuitive measure of the phase relationship between the two regions using circular–circular correlation. Figure 4b–c examines whether the activity peaks of the two regions are aligned across path directions using cross-frequency amplitude–phase coupling, given our hypothesis that the spatial phase of the HPC depends on EC projections. These two analyses are complementary: a phase correlation does not necessarily imply peak-to-peak alignment, and conversely, peak alignment does not always yield a statistically significant phase correlation. We therefore combined multiple analytical approaches as a cross-validation across methods, providing convergent evidence for robust EC–HPC coupling.

(17) Figure 5: Do these results hold when you estimate performance just based on “deviation from the goal to ending locations” (without taking path length into account)?

We thank the reviewer for this thoughtful suggestion. Following the reviewer’s advice, we re-estimated behavioral performance using the deviation between the goal and ending locations (i.e., error size) and path length independently. As shown in the new Figure S9, no significant periodicity was observed in error size ($p > 0.05$), whereas a robust 3-fold periodicity was found for path length ($p < 0.05$, corrected for multiple comparisons).

We employed two behavioral metrics, (1) path length and (2) error size, for complementary reasons. In our task, participants navigated using four discrete keys corresponding to the cardinal directions (north, south, east, and west). This design inherently induces a 4-fold bias in path directions, as described in the “Behavioral performance” subsection of the Methods. To minimize this artifact, we computed the objectively optimal path length and used it to calibrate participants’ path lengths. However, error size could not be corrected in the same manner and retained a residual 4-fold tendency (see Figure S9d).

Given that both path length and error size are behaviorally relevant and capture distinct aspects of task performance, we decided to retain both measures when quantifying behavioral periodicity. This clarification has been incorporated into the “Behavioral performance” subsection of the Methods, and the 2nd paragraph of the “3-fold periodicity in human behavior” subsection of the Results.

(18) Phase locking between behavioral performance and hippocampal activity: What is your way of creating surrogates here?

We thank the reviewer for this helpful question. Surrogate datasets were generated by circularly shifting the signal series along the direction axis across all possible offsets (following Canolty et al., 2006). This procedure preserves the internal phase structure within each domain while disrupting consistent phase alignment, thereby removing any systematic coupling between the two signals. Each surrogate dataset underwent identical filtering and coherence computation to generate a null distribution, and the observed coherence strength was compared with this distribution using paired t-tests across participants. The statistical analysis section has been systematically revised to incorporate these methodological details.

(19) I could not follow why the authors equate 3-fold symmetry with vectorial representations. This includes statements such as “these empirical findings provide a potential explanation for the formation of vectorial representation observed in the HPC.” Please clarify.

We thank the reviewer for raising this point. Please refer to our response to R2 for Reviewer #1 and the revised Introduction (paragraphs 2–4), where we explicitly explain why the three primary axes of the hexagonal grid cell code can manifest as vector fields.

(20) It was unclear whether the sentence “The EC provides a foundation for the formation of periodic representations in the HPC” is based on the authors’ observations or on other findings. If based on the authors’ findings, this statement seems too strong, given that no

other studies have reported periodic representations in the hippocampus to date (to the best of my knowledge).

We thank the reviewer for this comment. We agree that the original wording lacked sufficient rigor. We have extensively revised the 3rd paragraph of the Discussion section with more cautious language by reducing overinterpretation and emphasizing the consistency of our findings with prior empirical evidence, as follows: “The EC–HPC PhaseSync model demonstrates how a vectorial representation may emerge in the HPC from the projections of populations of periodic grid codes in the EC. The model was motivated by two observations. First, the EC intrinsically serves as the major source of hippocampal input (Witter and Amaral, 1991; van Groen et al., 2003; Garcia and Buffalo, 2020), and grid codes exhibit nearly invariant spatial orientations (Hafting et al., 2005; Gardner et al., 2022). Second, mental planning, characterized by “forward replay” (Dragoi and Tonegawa, 2011; Pfeiffer, 2020), has the capacity to activate populations of grid cells that represent sequential experiences in the absence of actual physical movement (Nyberg et al., 2022). We hypothesize that an integrated path code of sequential experiences may eventually be generated in the HPC, providing a vectorial gradient toward the goal location. The path code exhibits regular, vector-like representations when the path direction aligns with the orientations of grid axes, and becomes irregular when they misalign. This explanation is consistent with the band-like representations observed in the dorsomedial EC (Krupic et al., 2012) and the irregular activity fields of trace cells in the HPC (Poulter et al., 2021). ”

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